

PROBABILITY OF EVENTS

1.1. Introduction

During his lecture in 1929, Bertrand Russel said, “*Probability is the most important concept in modern science, especially as nobody has the slightest notion what it means.*” Most people have some vague ideas about what *probability of an event* means. The interpretation of the word *probability* involves synonyms such as chance, odds, uncertainty, prevalence, risk, expectancy etc. “*We use probability when we want to make an affirmation, but are not quite sure,*” writes J.R. Lucas.

There are many distinct interpretations of the word *probability*. A complete discussion of these interpretations will take us to areas such as philosophy, theory of algorithm and randomness, religion, etc. Thus, we will only focus on two extreme interpretations. One interpretation is due to the so-called objective school and the other is due to the subjective school.

The subjective school defines probabilities as subjective assignments based on rational thought with available information. Some subjective probabilists interpret probabilities as the degree of belief. Thus, it is difficult to interpret the probability of an event.

The objective school defines probabilities to be “*long run*” relative frequencies. This means that one should compute a probability by taking the number of favorable outcomes of an experiment and dividing it by total numbers of the possible outcomes of the experiment, and then taking the limit as the number of trials becomes large. Some statisticians object to the word “long run”. The philosopher and statistician John Keynes said “*in the long run we are all dead*”. The objective school uses the theory developed by

Von Mises (1928) and Kolmogorov (1965). The Russian mathematician Kolmogorov gave the solid foundation of probability theory using measure theory. The advantage of Kolmogorov's theory is that one can construct probabilities according to the rules, compute other probabilities using axioms, and then interpret these probabilities.

In this book, we will study mathematically one interpretation of probability out of many. In fact, we will study probability theory based on the theory developed by the late Kolmogorov. There are many applications of probability theory. We are studying probability theory because we would like to study mathematical statistics. Statistics is concerned with the development of methods and their applications for collecting, analyzing and interpreting quantitative data in such a way that the reliability of a conclusion based on data may be evaluated objectively by means of probability statements. Probability theory is used to evaluate the reliability of conclusions and inferences based on data. Thus, probability theory is fundamental to mathematical statistics.

For an event A of a discrete sample space S , the probability of A can be computed by using the formula

$$P(A) = \frac{N(A)}{N(S)}$$

where $N(A)$ denotes the number of elements of A and $N(S)$ denotes the number of elements in the sample space S . For a discrete case, the probability of an event A can be computed by counting the number of elements in A and dividing it by the number of elements in the sample space S .

Consider a standard deck of cards that has 26 red cards and 26 black cards. The probability of selecting a black card is $26/52 = 0.50$ because there are $X = 26$ black cards and $T = 52$ total cards. What does this probability mean? If each card is replaced after it is selected, does it mean that 1 out of the next 2 cards selected will be black? No, because you cannot say for certain what will happen on the next several selections. However, you can say that in the long run, if this selection process is continually repeated, the proportion of black cards selected will approach 0.50.

A standard six-sided die has six faces. Each face of the die contains either one, two, three, four, five, or six dots. If you roll a die, what is the probability that you will get a face with five dots?

SOLUTION Each face is equally likely to occur. Because there are six faces, the probability of getting a face with five dots is $1/6$.

There are three types of probability:

- *A priori*
- Empirical
- Subjective

The preceding examples use the *a priori* probability approach because the number of ways the event occurs and the total number of possible outcomes are known from the composition of the deck of cards or the faces of the die.

In the **empirical probability** approach, the probabilities are based on observed data, not on prior knowledge of a process. Surveys are often used to generate empirical probabilities. Examples of this type of probability are the proportion of individuals in the Using Statistics scenario who actually purchase big-screen televisions, the proportion of registered voters who prefer a certain political candidate, and the proportion of students who have part-time jobs. For example, if you take a survey of students, and 60% state that they have part-time jobs, then there is a 0.60 probability that an individual student has a part-time job.

The third approach to probability, **subjective probability**, differs from the other two approaches because subjective probability differs from person to person. For example, the development team for a new product may assign a probability of 0.60 to the chance of success for the product, while the president of the company may be less optimistic and assign a probability of 0.30. The assignment of subjective probabilities to various outcomes is usually based on a combination of an individual's past experience, personal opinion, and analysis of a particular situation. Subjective probability is especially useful in making decisions in situations in which you cannot use *a priori* probability or empirical probability.

In the next section, we develop various counting techniques. The branch of mathematics that deals with the various counting techniques is called combinatorics.

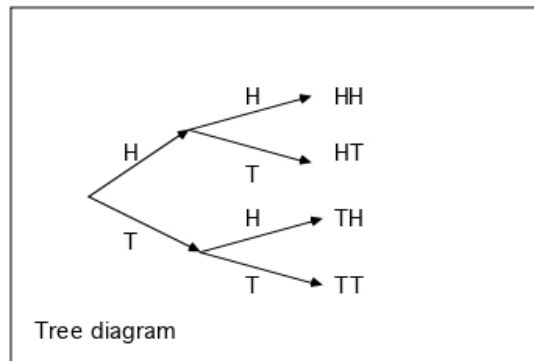
1.2. Counting Techniques

There are three basic counting techniques. They are multiplication rule, permutation and combination.

1.2.1 Multiplication Rule. If E_1 is an experiment with n_1 outcomes and E_2 is an experiment with n_2 possible outcomes, then the experiment which consists of performing E_1 first and then E_2 consists of $n_1 n_2$ possible outcomes.

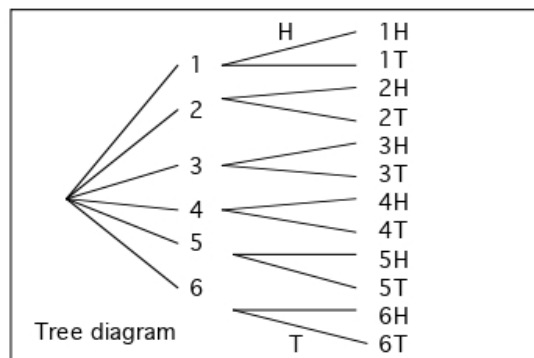
Example 1.1. Find the possible number of outcomes in a sequence of two tosses of a fair coin.

Answer: The number of possible outcomes is $2 \cdot 2 = 4$. This is evident from the following tree diagram.



Example 1.2. Find the number of possible outcomes of the rolling of a die and then tossing a coin.

Answer: Here $n_1 = 6$ and $n_2 = 2$. Thus by multiplication rule, the number of possible outcomes is 12.



Example 1.3. How many different license plates are possible if Kentucky uses three letters followed by three digits.

Answer:

$$\begin{aligned}
 & (26)^3 (10)^3 \\
 & = (17576) (1000) \\
 & = 17,576,000.
 \end{aligned}$$

1.2.2. Permutation

Consider a set of 4 objects. Suppose we want to fill 3 positions with objects selected from the above 4. Then the number of possible ordered arrangements is 24 and they are

a b c	b a c	c a b	d a b
a b d	b a d	c a d	d a c
a c b	b c a	c b a	d b c
a c d	b c d	c b d	d b a
a d c	b d a	c d b	d c a
a d b	b d c	c d a	d c b

The number of possible ordered arrangements can be computed as follows: Since there are 3 positions and 4 objects, the first position can be filled in 4 different ways. Once the first position is filled the remaining 2 positions can be filled from the remaining 3 objects. Thus, the second position can be filled in 3 ways. The third position can be filled in 2 ways. Then the total number of ways 3 positions can be filled out of 4 objects is given by

$$(4)(3)(2) = 24.$$

In general, if r positions are to be filled from n objects, then the total number of possible ways they can be filled are given by

$$\begin{aligned} & n(n-1)(n-2) \cdots (n-r+1) \\ &= \frac{n!}{(n-r)!} \\ &= {}_nP_r. \end{aligned}$$

Thus, ${}_nP_r$ represents the number of ways r positions can be filled from n objects.

Definition 1.1. Each of the ${}_nP_r$ arrangements is called a permutation of n objects taken r at a time.

Example 1.4. How many permutations are there of all three of letters a, b, and c?

Answer:

$$\begin{aligned} {}_3P_3 &= \frac{n!}{(n-r)!} \\ &= \frac{3!}{0!} = 6 \end{aligned}$$

Example 1.5. Find the number of permutations of n distinct objects.

Answer:

$${}_nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!.$$

Example 1.6. Four names are drawn from the 24 members of a club for the offices of President, Vice-President, Treasurer, and Secretary. In how many different ways can this be done?

Answer:

$$\begin{aligned} {}_{24}P_4 &= \frac{(24)!}{(20)!} \\ &= (24)(23)(22)(21) \\ &= 255,024. \end{aligned}$$

1.2.3. Combination

In permutation, order is important. But in many problems the order of selection is not important and interest centers only on the set of r objects.

Let c denote the number of subsets of size r that can be selected from n different objects. The r objects in each set can be ordered in ${}_rP_r$ ways. Thus we have

$${}_nP_r = c ({}_rP_r).$$

From this, we get

$$c = \frac{{}_nP_r}{{}_rP_r} = \frac{n!}{(n-r)!r!}$$

The number c is denoted by $\binom{n}{r}$. Thus, the above can be written as

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}.$$

Definition 1.2. Each of the $\binom{n}{r}$ unordered subsets is called a combination of n objects taken r at a time.

Example 1.7. How many committees of two chemists and one physicist can be formed from 4 chemists and 3 physicists?

Answer:

$$\begin{aligned}\binom{4}{2} \binom{3}{1} \\ &= (6) (3) \\ &= 18.\end{aligned}$$

Thus 18 different committees can be formed.

1.2.4. Binomial Theorem

We know from lower level mathematics courses that

$$\begin{aligned}(x + y)^2 &= x^2 + 2xy + y^2 \\ &= \binom{2}{0} x^2 + \binom{2}{1} xy + \binom{2}{2} y^2 \\ &= \sum_{k=0}^2 \binom{2}{k} x^{2-k} y^k.\end{aligned}$$

Similarly

$$\begin{aligned}(x + y)^3 &= x^3 + 3x^2y + 3xy^2 + y^3 \\ &= \binom{3}{0} x^3 + \binom{3}{1} x^2y + \binom{3}{2} xy^2 + \binom{3}{3} y^3 \\ &= \sum_{k=0}^3 \binom{3}{k} x^{3-k} y^k.\end{aligned}$$

In general, using induction arguments, we can show that

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k.$$

This result is called the Binomial Theorem. The coefficient $\binom{n}{k}$ is called the binomial coefficient. A combinatorial proof of the Binomial Theorem follows. If we write $(x + y)^n$ as the n times the product of the factor $(x + y)$, that is

$$(x + y)^n = (x + y) (x + y) (x + y) \cdots (x + y),$$

then the coefficient of $x^{n-k}y^k$ is $\binom{n}{k}$, that is the number of ways in which we can choose the k factors providing the y 's.

Remark 1.1. In 1665, Newton discovered the Binomial Series. The Binomial Series is given by

$$\begin{aligned}(1+y)^\alpha &= 1 + \binom{\alpha}{1}y + \binom{\alpha}{2}y^2 + \cdots + \binom{\alpha}{n}y^n + \cdots \\ &= 1 + \sum_{k=1}^{\infty} \binom{\alpha}{k}y^k,\end{aligned}$$

where α is a real number and

$$\binom{\alpha}{k} = \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-k+1)}{k!}.$$

This $\binom{\alpha}{k}$ is called the generalized binomial coefficient.

Now, we investigate some properties of the binomial coefficients.

Theorem 1.1. Let $n \in \mathbf{N}$ (the set of natural numbers) and $r = 0, 1, 2, \dots, n$. Then

$$\binom{n}{r} = \binom{n}{n-r}.$$

Proof: By direct verification, we get

$$\begin{aligned}\binom{n}{n-r} &= \frac{n!}{(n-n+r)!(n-r)!} \\ &= \frac{n!}{r!(n-r)!} \\ &= \binom{n}{r}.\end{aligned}$$

This theorem says that the binomial coefficients are symmetrical.

Example 1.8. Evaluate $\binom{3}{1} + \binom{3}{2} + \binom{3}{0}$.

Answer: Since the combinations of 3 things taken 1 at a time are 3, we get $\binom{3}{1} = 3$. Similarly, $\binom{3}{0}$ is 1. By Theorem 1,

$$\binom{3}{1} = \binom{3}{2} = 3.$$

Hence

$$\binom{3}{1} + \binom{3}{2} + \binom{3}{0} = 3 + 3 + 1 = 7.$$

Theorem 1.2. For any positive integer n and $r = 1, 2, 3, \dots, n$, we have

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}.$$

Proof:

$$\begin{aligned} (1+y)^n &= (1+y)(1+y)^{n-1} \\ &= (1+y)^{n-1} + y(1+y)^{n-1} \\ \sum_{r=0}^n \binom{n}{r} y^r &= \sum_{r=0}^{n-1} \binom{n-1}{r} y^r + y \sum_{r=0}^{n-1} \binom{n-1}{r} y^r \\ &= \sum_{r=0}^{n-1} \binom{n-1}{r} y^r + \sum_{r=0}^{n-1} \binom{n-1}{r} y^{r+1}. \end{aligned}$$

Equating the coefficients of y^r from both sides of the above expression, we obtain

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$$

and the proof is now complete.

Example 1.9. Evaluate $\binom{23}{10} + \binom{23}{9} + \binom{24}{11}$.

Answer:

$$\begin{aligned} &\binom{23}{10} + \binom{23}{9} + \binom{24}{11} \\ &= \binom{24}{10} + \binom{24}{11} \\ &= \binom{25}{11} \\ &= \frac{25!}{(14)!(11)!} \\ &= 4,457,400. \end{aligned}$$

Example 1.10. Use the Binomial Theorem to show that $\sum_{r=0}^n (-1)^r \binom{n}{r} = 0$.

Answer: Using the Binomial Theorem, we get

$$(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$$

for all real numbers x . Letting $x = -1$ in the above, we get

$$0 = \sum_{r=0}^n \binom{n}{r} (-1)^r.$$

Theorem 1.3. Let m and n be positive integers. Then

$$\sum_{r=0}^k \binom{m}{r} \binom{n}{k-r} = \binom{m+n}{k}.$$

Proof:

$$(1+y)^{m+n} = (1+y)^m (1+y)^n$$

$$\sum_{r=0}^{m+n} \binom{m+n}{r} y^r = \left\{ \sum_{r=0}^m \binom{m}{r} y^r \right\} \left\{ \sum_{r=0}^n \binom{n}{r} y^r \right\}.$$

Equating the coefficients of y^k from the both sides of the above expression, we obtain

$$\binom{m+n}{k} = \binom{m}{0} \binom{n}{k} + \binom{m}{1} \binom{n}{k-1} + \cdots + \binom{m}{k} \binom{n}{k-k}$$

and the conclusion of the theorem follows.

Example 1.11. Show that

$$\sum_{r=0}^n \binom{n}{r}^2 = \binom{2n}{n}.$$

Answer: Let $k = n$ and $m = n$. Then from Theorem 3, we get

$$\sum_{r=0}^k \binom{m}{r} \binom{n}{k-r} = \binom{m+n}{k}$$

$$\sum_{r=0}^n \binom{n}{r} \binom{n}{n-r} = \binom{2n}{n}$$

$$\sum_{r=0}^n \binom{n}{r} \binom{n}{r} = \binom{2n}{n}$$

$$\sum_{r=0}^n \binom{n}{r}^2 = \binom{2n}{n}.$$

Theorem 1.4. Let n be a positive integer and $k = 1, 2, 3, \dots, n$. Then

$$\binom{n}{k} = \sum_{m=k-1}^{n-1} \binom{m}{k-1}.$$

Proof: In order to establish the above identity, we use the Binomial Theorem together with the following result of the elementary algebra

$$x^n - y^n = (x - y) \sum_{k=0}^{n-1} x^k y^{n-1-k}.$$

Note that

$$\begin{aligned} \sum_{k=1}^n \binom{n}{k} x^k &= \sum_{k=0}^n \binom{n}{k} x^k - 1 \\ &= (x+1)^n - 1^n \quad \text{by Binomial Theorem} \\ &= (x+1-1) \sum_{m=0}^{n-1} (x+1)^m \quad \text{by above identity} \\ &= x \sum_{m=0}^{n-1} \sum_{j=0}^m \binom{m}{j} x^j \\ &= \sum_{m=0}^{n-1} \sum_{j=0}^m \binom{m}{j} x^{j+1} \\ &= \sum_{k=1}^n \sum_{m=k-1}^{n-1} \binom{m}{k-1} x^k. \end{aligned}$$

Hence equating the coefficient of x^k , we obtain

$$\binom{n}{k} = \sum_{m=k-1}^{n-1} \binom{m}{k-1}.$$

This completes the proof of the theorem.

The following result

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{n_1+n_2+\dots+n_m=n} \binom{n}{n_1, n_2, \dots, n_m} x_1^{n_1} x_2^{n_2} \dots x_m^{n_m}$$

is known as the multinomial theorem and it generalizes the binomial theorem. The sum is taken over all positive integers $n_1, n_2, \dots, n_m$ such that $n_1 + n_2 + \dots + n_m = n$, and

$$\binom{n}{n_1, n_2, \dots, n_m} = \frac{n!}{n_1! n_2! \dots n_m!}.$$

This coefficient is known as the multinomial coefficient.

1.3. Probability Measure

A random experiment is an experiment whose outcomes cannot be predicted with certainty. However, in most cases the collection of every possible outcome of a random experiment can be listed.

Definition 1.3. A sample space of a random experiment is the collection of all possible outcomes.

Example 1.12. What is the sample space for an experiment in which we select a rat at random from a cage and determine its sex?

Answer: The sample space of this experiment is

$$S = \{M, F\}$$

where M denotes the male rat and F denotes the female rat.

Example 1.13. What is the sample space for an experiment in which the state of Kentucky picks a three digit integer at random for its daily lottery?

Answer: The sample space of this experiment is

$$S = \{000, 001, 002, \dots, 998, 999\}.$$

Example 1.14. What is the sample space for an experiment in which we roll a pair of dice, one red and one green?

Answer: The sample space S for this experiment is given by

$$S = \begin{matrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{matrix}$$

This set S can be written as

$$S = \{(x, y) \mid 1 \leq x \leq 6, 1 \leq y \leq 6\}$$

where x represents the number rolled on red die and y denotes the number rolled on green die.

Definition 1.4. Each element of the sample space is called a sample point.

Definition 1.5. If the sample space consists of a countable number of sample points, then the sample space is said to be a countable sample space.

Definition 1.6. If a sample space contains an uncountable number of sample points, then it is called a continuous sample space.

In some tutorials the sample space is denoted by Ω .

Example 1. If the outcome of an experiment consists of the determination of the sex of a newborn child, then

$$\Omega = \{\omega_1, \omega_2\},$$

where the outcome ω_1 means that the child is a girl and ω_2 that it is a boy.

Example 2. If an experiment consists of tossing a coin on a smooth surface then Ω is the same as in Example 1, that is

$$\Omega = \{\omega_1, \omega_2\},$$

where ω_1 corresponds to a head and ω_2 corresponds to a tail.

Example 3. If we toss a symmetrical (or a fair) die on a smooth surface then

$$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4, \omega_5, \omega_6\},$$

where ω_i corresponds to the outcome when on the opened lateral face appear i points, $i = 1, 2, \dots, 6$.

Example 4. Now an experiment consists of tossing a coin twice. In this case we take as the set of possible outcomes (i.e. as the sample space) the 4-element set

$$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\},$$

where the outcomes are the following

$\omega_1 = hh$ is the outcome: heads on the first toss and heads on the second;

$\omega_2 = ht$ is the outcome: heads on the first toss and tails on the second;

$\omega_3 = th$ is the outcome: tails on the first toss and heads on the second;

$\omega_4 = tt$ is the outcome: tails on the first toss and tails on the second.

Example 5. Now two symmetrical dice are thrown on a smooth surface then

$$\Omega = \{\omega_1 = (1, 1), \quad \omega_2 = (1, 2), \quad \omega_3 = (2, 1), \dots, \quad \omega_{36} = (6, 6)\},$$

where (i, j) is the outcome that the first die equals i and the second die equals j , i. e. Ω is a 36 element set.

Example 6. A card is selected at random from a deck of 52 playing cards, then Ω is 52 element set.

Remark 1. It should be pointed out that the sample space of an experiment can be defined in more than one way. Observers with different conceptions of what could possibly be observed will arrive at different sample spaces. For instance, in Example 2, the sample space Ω might consist of three elements, if we desired to include the possibility that the coin might stand on its edge or rim. Then

$$\Omega = \{\omega_1, \omega_2, \omega_3\},$$

where ω_1, ω_2 are as in Example 2 and the outcome ω_3 represents the possibility of the coin standing on its rim.

There is yet a fourth possibility; the coin might be lost by rolling away when it lands. The sample space Ω would then be

$$\Omega = \{\omega_1, \omega_2, \omega_3, \omega_4\}$$

in which the outcome ω_4 denotes the possibility of loss.

In these examples Ω is finite. However, there exist experiments for which the number of elements of Ω is infinite.

Example 7. Assume we toss a coin until a tail opens. Then

$$\Omega = \{\omega_1, \omega_2, \dots, \omega_n, \dots\},$$

where ω_i corresponds to the outcome for which a coin rolls exactly i times, i. e. we will have to toss the coin i times until a tail opens.

Example 8. If we choose a point at random in the bounded subset D of n -dimensional Euclidean space $\mathbb{R}^n$ then

$$\Omega = D.$$

In Example 7 Ω is a countable set and in Example 8 Ω is a continuum set.

In general, a sample space is said to be *discrete* if it has finitely many or a countable infinity of elements. If the points (outcomes) of a sample space constitute a continuum, for example, all the points on a line, all the points on a line segment, or all the points in a plane, the sample space is said to be *continuous*.

An event A is a subset of the sample space S . It seems obvious that if A and B are events in sample space S , then $A \cup B$, A^c , $A \cap B$ are also entitled to be events. Thus precisely we define an event as follows:

Definition 1.7. A subset A of the sample space S is said to be an event if it belongs to a collection $\mathcal{F}$ of subsets of S satisfying the following three rules: (a) $S \in \mathcal{F}$; (b) if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$; and (c) if $A_j \in \mathcal{F}$ for $j \geq 1$, then $\bigcup_{j=1}^{\infty} A_j \in \mathcal{F}$. The collection $\mathcal{F}$ is called an event space or a σ -field. If A is the outcome of an experiment, then we say that the event A has occurred.

Example 1.15. Describe the sample space of rolling a die and interpret the event $\{1, 2\}$.

Answer: The sample space of this experiment is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

The event $\{1, 2\}$ means getting either a 1 or a 2.

Example 1.16. First describe the sample space of rolling a pair of dice, then describe the event A that the sum of numbers rolled is 7.

Answer: The sample space of this experiment is

$$S = \{(x, y) \mid x, y = 1, 2, 3, 4, 5, 6\}$$

and

$$A = \{(1, 6), (6, 1), (2, 5), (5, 2), (4, 3), (3, 4)\}.$$

Definition 1.8. Let S be the sample space of a random experiment. A probability measure $P : \mathcal{F} \rightarrow [0, 1]$ is a set function which assigns real numbers to the various events of S satisfying

- (P1) $P(A) \geq 0$ for all event $A \in \mathcal{F}$,
- (P2) $P(S) = 1$,

$$(P3) \quad P\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} P(A_k)$$

if $A_1, A_2, A_3, \dots, A_k, \dots$ are mutually disjoint events of S .

Any set function with the above three properties is a probability measure for S . For a given sample space S , there may be more than one probability measure. The probability of an event A is the value of the probability measure at A , that is

$$Prob(A) = P(A).$$

Theorem 1.5. If $\emptyset$ is a empty set (that is an impossible event), then

$$P(\emptyset) = 0.$$

Proof: Let $A_1 = S$ and $A_i = \emptyset$ for $i = 2, 3, \dots, \infty$. Then

$$S = \bigcup_{i=1}^{\infty} A_i$$

where $A_i \cap A_j = \emptyset$ for $i \neq j$. By axiom 2 and axiom 3, we get

$$\begin{aligned} 1 &= P(S) \quad (\text{by axiom 2}) \\ &= P\left(\bigcup_{i=1}^{\infty} A_i\right) \\ &= \sum_{i=1}^{\infty} P(A_i) \quad (\text{by axiom 3}) \\ &= P(A_1) + \sum_{i=2}^{\infty} P(A_i) \\ &= P(S) + \sum_{i=2}^{\infty} P(\emptyset) \\ &= 1 + \sum_{i=2}^{\infty} P(\emptyset). \end{aligned}$$

Therefore

$$\sum_{i=2}^{\infty} P(\emptyset) = 0.$$

Since $P(\emptyset) \geq 0$ by axiom 1, we have

$$P(\emptyset) = 0$$

and the proof of the theorem is complete.

This theorem says that the probability of an impossible event is zero. Note that if the probability of an event is zero, that does not mean the event is empty (or impossible). There are random experiments in which there are infinitely many events each with probability 0. Similarly, if A is an event with probability 1, then it does not mean A is the sample space S . In fact there are random experiments in which one can find infinitely many events each with probability 1.

Theorem 1.6. Let $\{A_1, A_2, \dots, A_n\}$ be a finite collection of n events such that $A_i \cap A_j = \emptyset$ for $i \neq j$. Then

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n P(A_i).$$

Proof: Consider the collection $\{A'_i\}_{i=1}^{\infty}$ of the subsets of the sample space S such that

$$A'_1 = A_1, A'_2 = A_2, \dots, A'_n = A_n$$

and

$$A'_{n+1} = A'_{n+2} = A'_{n+3} = \dots = \emptyset.$$

Hence

$$\begin{aligned} P\left(\bigcup_{i=1}^n A_i\right) &= P\left(\bigcup_{i=1}^{\infty} A'_i\right) \\ &= \sum_{i=1}^{\infty} P(A'_i) \\ &= \sum_{i=1}^n P(A'_i) + \sum_{i=n+1}^{\infty} P(A'_i) \\ &= \sum_{i=1}^n P(A_i) + \sum_{i=n+1}^{\infty} P(\emptyset) \\ &= \sum_{i=1}^n P(A_i) + 0 \\ &= \sum_{i=1}^n P(A_i) \end{aligned}$$

and the proof of the theorem is now complete.

When $n = 2$, the above theorem yields $P(A_1 \cup A_2) = P(A_1) + P(A_2)$ where A_1 and A_2 are disjoint (or mutually exclusive) events.

In the following theorem, we give a method for computing probability of an event A by knowing the probabilities of the elementary events of the sample space S .

Theorem 1.7. If A is an event of a discrete sample space S , then the probability of A is equal to the sum of the probabilities of its elementary events.

Proof: Any set A in S can be written as the union of its singleton sets. Let $\{O_i\}_{i=1}^{\infty}$ be the collection of all the singleton sets (or the elementary events) of A . Then

$$A = \bigcup_{i=1}^{\infty} O_i.$$

By axiom (P3), we get

$$\begin{aligned} P(A) &= P\left(\bigcup_{i=1}^{\infty} O_i\right) \\ &= \sum_{i=1}^{\infty} P(O_i). \end{aligned}$$

Example 1.17. If a fair coin is tossed twice, what is the probability of getting at least one head?

Answer: The sample space of this experiment is

$$S = \{HH, HT, TH, TT\}.$$

The event A is given by

$$\begin{aligned} A &= \{\text{at least one head}\} \\ &= \{HH, HT, TH\}. \end{aligned}$$

By Theorem 1.7, the probability of A is the sum of the probabilities of its elementary events. Thus, we get

$$\begin{aligned} P(A) &= P(HH) + P(HT) + P(TH) \\ &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \\ &= \frac{3}{4}. \end{aligned}$$

Remark 1.2. Notice that here we are not computing the probability of the elementary events by taking the number of points in the elementary event and dividing by the total number of points in the sample space. We are using the randomness to obtain the probability of the elementary events. That is, we are assuming that each outcome is equally likely. This is why the randomness is an integral part of probability theory.

Corollary 1.1. If S is a finite sample space with n sample elements and A is an event in S with m elements, then the probability of A is given by

$$P(A) = \frac{m}{n}.$$

Proof: By the previous theorem, we get

$$\begin{aligned} P(A) &= P\left(\bigcup_{i=1}^m O_i\right) \\ &= \sum_{i=1}^m P(O_i) \\ &= \sum_{i=1}^m \frac{1}{n} \\ &= \frac{m}{n}. \end{aligned}$$

The proof is now complete.

Example 1.18. A die is loaded in such a way that the probability of the face with j dots turning up is proportional to j for $j = 1, 2, \dots, 6$. What is the probability, in one roll of the die, that an odd number of dots will turn up?

Answer:

$$\begin{aligned} P(\{j\}) &\propto j \\ &= k j \end{aligned}$$

where k is a constant of proportionality. Next, we determine this constant k by using the axiom (P2). Using Theorem 1.5, we get

$$\begin{aligned} P(S) &= P(\{1\}) + P(\{2\}) + P(\{3\}) + P(\{4\}) + P(\{5\}) + P(\{6\}) \\ &= k + 2k + 3k + 4k + 5k + 6k \\ &= (1 + 2 + 3 + 4 + 5 + 6) k \\ &= \frac{(6)(6+1)}{2} k \\ &= 21k. \end{aligned}$$

Using (P2), we get

$$21k = 1.$$

Thus $k = \frac{1}{21}$. Hence, we have

$$P(\{j\}) = \frac{j}{21}.$$

Now, we want to find the probability of the odd number of dots turning up.

$$\begin{aligned} P(\text{odd numbered dot will turn up}) &= P(\{1\}) + P(\{3\}) + P(\{5\}) \\ &= \frac{1}{21} + \frac{3}{21} + \frac{5}{21} \\ &= \frac{9}{21}. \end{aligned}$$

Remark 1.3. Recall that the sum of the first n integers is equal to $\frac{n}{2}(n+1)$. That is,

$$1 + 2 + 3 + \cdots + (n-2) + (n-1) + n = \frac{n(n+1)}{2}.$$

This formula was first proven by Gauss (1777-1855) when he was a young school boy.

Remark 1.4. Gauss proved that the sum of the first n positive integers is $n\frac{(n+1)}{2}$ when he was a school boy. Kolmogorov, the father of modern probability theory, proved that the sum of the first n odd positive integers is n^2 , when he was five years old.

1.4. Some Properties of the Probability Measure

Next, we present some theorems that will illustrate the various intuitive properties of a probability measure.

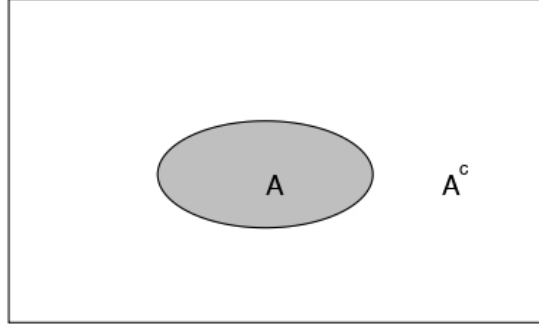
Theorem 1.8. If A be any event of the sample space S , then

$$P(A^c) = 1 - P(A)$$

where A^c denotes the complement of A with respect to S .

Proof: Let A be any subset of S . Then $S = A \cup A^c$. Further A and A^c are mutually disjoint. Thus, using (P3), we get

$$\begin{aligned} 1 = P(S) &= P(A \cup A^c) \\ &= P(A) + P(A^c). \end{aligned}$$



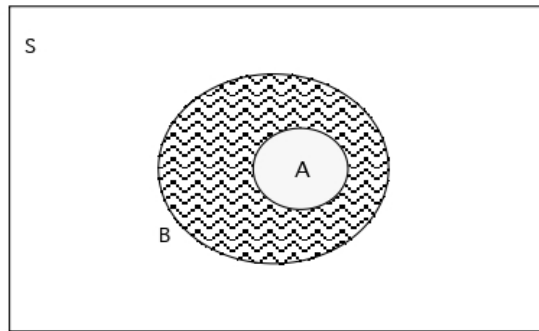
Hence, we see that

$$P(A^c) = 1 - P(A).$$

This completes the proof.

Theorem 1.9. If $A \subseteq B \subseteq S$, then

$$P(A) \leq P(B).$$



Proof: Note that $B = A \cup (B \setminus A)$ where $B \setminus A$ denotes all the elements x that are in B but not in A . Further, $A \cap (B \setminus A) = \emptyset$. Hence by (P3), we get

$$\begin{aligned} P(B) &= P(A \cup (B \setminus A)) \\ &= P(A) + P(B \setminus A). \end{aligned}$$

By axiom (P1), we know that $P(B \setminus A) \geq 0$. Thus, from the above, we get

$$P(B) \geq P(A)$$

and the proof is complete.

Theorem 1.10. If A is any event in S , then

$$0 \leq P(A) \leq 1.$$

Proof: Follows from axioms (P1) and (P2) and Theorem 1.8.

Theorem 1.10. If A and B are any two events, then

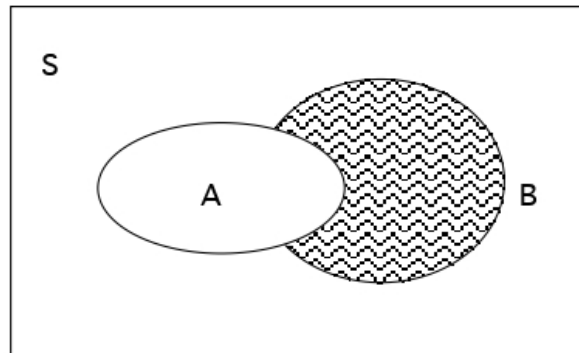
$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof: It is easy to see that

$$A \cup B = A \cup (A^c \cap B)$$

and

$$A \cap (A^c \cap B) = \emptyset.$$

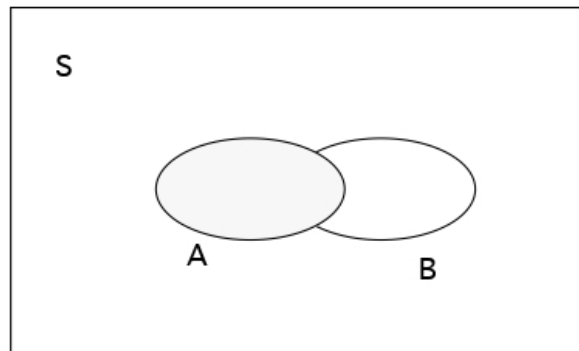


Hence by (P3), we get

$$P(A \cup B) = P(A) + P(A^c \cap B) \tag{1.1}$$

But the set B can also be written as

$$B = (A \cap B) \cup (A^c \cap B)$$



Therefore, by (P3), we get

$$P(B) = P(A \cap B) + P(A^c \cap B). \quad (1.2)$$

Eliminating $P(A^c \cap B)$ from (1.1) and (1.2), we get

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

and the proof of the theorem is now complete.

This above theorem tells us how to calculate the probability that at least one of A and B occurs.

Example 1.19. If $P(A) = 0.25$ and $P(B) = 0.8$, then show that $0.05 \leq P(A \cap B) \leq 0.25$.

Answer: Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, by Theorem 1.8, we get

$$P(A \cap B) \leq P(A) \quad \text{and also} \quad P(A \cap B) \leq P(B).$$

Hence

$$P(A \cap B) \leq \min\{P(A), P(B)\}.$$

This shows that

$$P(A \cap B) \leq 0.25. \quad (1.3)$$

Since $A \cup B \subseteq S$, by Theorem 1.8, we get

$$P(A \cup B) \leq P(S)$$

That is, by Theorem 1.10

$$P(A) + P(B) - P(A \cap B) \leq P(S).$$

Hence, we obtain

$$0.8 + 0.25 - P(A \cap B) \leq 1$$

and this yields

$$0.8 + 0.25 - 1 \leq P(A \cap B).$$

From this, we get

$$0.05 \leq P(A \cap B). \quad (1.4)$$

From (1.3) and (1.4), we get

$$0.05 \leq P(A \cap B) \leq 0.25.$$

Example 1.20. Let A and B be events in a sample space S such that $P(A) = \frac{1}{2} = P(B)$ and $P(A^c \cap B^c) = \frac{1}{3}$. Find $P(A \cup B^c)$.

Answer: Notice that

$$A \cup B^c = A \cup (A^c \cap B^c).$$

Hence,

$$\begin{aligned} P(A \cup B^c) &= P(A) + P(A^c \cap B^c) \\ &= \frac{1}{2} + \frac{1}{3} \\ &= \frac{5}{6}. \end{aligned}$$

Theorem 1.11. If A_1 and A_2 are two events such that $A_1 \subseteq A_2$, then

$$P(A_2 \setminus A_1) = P(A_2) - P(A_1).$$

Proof: The event A_2 can be written as

$$A_2 = A_1 \bigcup (A_2 \setminus A_1)$$

where the sets A_1 and $A_2 \setminus A_1$ are disjoint. Hence

$$P(A_2) = P(A_1) + P(A_2 \setminus A_1)$$

which is

$$P(A_2 \setminus A_1) = P(A_2) - P(A_1)$$

and the proof of the theorem is now complete.

Theorem 1.12. If $A_1, A_2, \dots, A_n, \dots$ is a sequence of events in sample space S such that $A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq \dots$, then

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n).$$

Similarly, if $B_1, B_2, \dots, B_n, \dots$ is a sequence of events in sample space S such that $B_1 \supseteq B_2 \supseteq \dots \supseteq B_n \supseteq \dots$, then

$$P\left(\bigcap_{n=1}^{\infty} B_n\right) = \lim_{n \rightarrow \infty} P(B_n).$$

Proof: Given an increasing sequence of events

$$A_1 \subseteq A_2 \subseteq \dots \subseteq A_n \subseteq \dots$$

we define a disjoint collection of events as follows:

$$\begin{aligned} E_1 &= A_1 \\ E_n &= A_n \setminus A_{n-1} \quad \forall n \geq 2. \end{aligned}$$

Then $\{E_n\}_{n=1}^{\infty}$ is a disjoint collection of events such that

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} E_n.$$

Further

$$\begin{aligned} P\left(\bigcup_{n=1}^{\infty} A_n\right) &= P\left(\bigcup_{n=1}^{\infty} E_n\right) \\ &= \sum_{n=1}^{\infty} P(E_n) \\ &= \lim_{m \rightarrow \infty} \sum_{n=1}^m P(E_n) \\ &= \lim_{m \rightarrow \infty} \left[P(A_1) + \sum_{n=2}^m [P(A_n) - P(A_{n-1})] \right] \\ &= \lim_{m \rightarrow \infty} P(A_m) \\ &= \lim_{n \rightarrow \infty} P(A_n). \end{aligned}$$

The second part of the theorem can be proved similarly.

Note that

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

and

$$\lim_{n \rightarrow \infty} B_n = \bigcap_{n=1}^{\infty} B_n.$$

Hence the results above theorem can be written as

$$P\left(\lim_{n \rightarrow \infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n)$$

and

$$P\left(\lim_{n \rightarrow \infty} B_n\right) = \lim_{n \rightarrow \infty} P(B_n)$$

and the Theorem 1.12 is called the continuity theorem for the probability measure.

GEOMETRIC PROBABILITIES

Let Ω be the same as in Example 8 i. e. we choose a point at random in the bounded subset D of n -dimensional Euclidean space $\mathbf{R}^n$. Therefore $\Omega = D$ and events are subsets of D . We suppose that volume of D does not equal 0, $V(D) \neq 0$ (V stands for volume). We define $P(A)$ by the following formula

$$P(A) = \frac{V(A)}{V(\Omega)}.$$

In particular, when $n = 2$

$$P(A) = \frac{S(A)}{S(\Omega)},$$

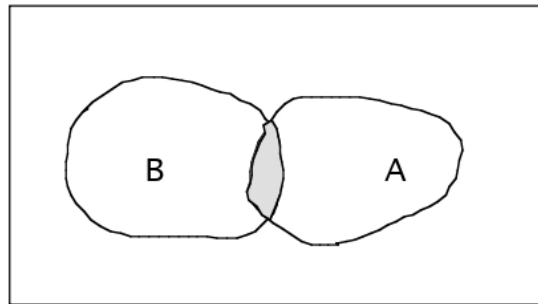
in which S stands for area.

CONDITIONAL PROBABILITIES AND BAYES' THEOREM

2.1. Conditional Probabilities

First, we give a heuristic argument for the definition of conditional probability, and then based on our heuristic argument, we define the conditional probability.

Consider a random experiment whose sample space is S . Let $B \subset S$. In many situations, we are only concerned with those outcomes that are elements of B . This means that we consider B to be our new sample space.



For the time being, suppose S is a nonempty finite sample space and B is a nonempty subset of S . Given this new discrete sample space B , how do we define the probability of an event A ? Intuitively, one should define the probability of A with respect to the new sample space B as (see the figure above)

$$P(A \text{ given } B) = \frac{\text{the number of elements in } A \cap B}{\text{the number of elements in } B}.$$

We denote the conditional probability of A given the new sample space B as $P(A/B)$. Hence with this notation, we say that

$$\begin{aligned} P(A/B) &= \frac{N(A \cap B)}{N(B)} \\ &= \frac{P(A \cap B)}{P(B)}, \end{aligned}$$

since $N(S) \neq 0$. Here $N(S)$ denotes the number of elements in S .

Thus, if the sample space is finite, then the above definition of the probability of an event A given that the event B has occurred makes sense intuitively. Now we define the conditional probability for any sample space (discrete or continuous) as follows.

Definition 2.1. Let S be a sample space associated with a random experiment. The conditional probability of an event A , given that event B has occurred, is defined by

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) > 0$.

This conditional probability measure $P(A/B)$ satisfies all three axioms of a probability measure. That is,

(CP1) $P(A/B) \geq 0$ for all event A

(CP2) $P(B/B) = 1$

(CP3) If $A_1, A_2, \dots, A_k, \dots$ are mutually exclusive events, then

$$P\left(\bigcup_{k=1}^{\infty} A_k/B\right) = \sum_{k=1}^{\infty} P(A_k/B).$$

Thus, it is a probability measure with respect to the new sample space B .

Example 2.1. A drawer contains 4 black, 6 brown, and 8 olive socks. Two socks are selected at random from the drawer. (a) What is the probability that both socks are of the same color? (b) What is the probability that both socks are olive if it is known that they are of the same color?

Answer: The sample space of this experiment consists of

$$S = \{(x, y) \mid x, y \in Bl, Ol, Br\}.$$

The cardinality of S is

$$N(S) = \binom{18}{2} = 153.$$

Let A be the event that two socks selected at random are of the same color. Then the cardinality of A is given by

$$\begin{aligned} N(A) &= \binom{4}{2} + \binom{6}{2} + \binom{8}{2} \\ &= 6 + 15 + 28 \\ &= 49. \end{aligned}$$

Therefore, the probability of A is given by

$$P(A) = \frac{49}{\binom{18}{2}} = \frac{49}{153}.$$

Let B be the event that two socks selected at random are olive. Then the cardinality of B is given by

$$N(B) = \binom{8}{2}$$

and hence

$$P(B) = \frac{\binom{8}{2}}{\binom{18}{2}} = \frac{28}{153}.$$

Notice that $B \subset A$. Hence,

$$\begin{aligned} P(B/A) &= \frac{P(A \cap B)}{P(A)} \\ &= \frac{P(B)}{P(A)} \\ &= \left(\frac{28}{153} \right) \left(\frac{153}{49} \right) \\ &= \frac{28}{49} = \frac{4}{7}. \end{aligned}$$

Let A and B be two mutually disjoint events in a sample space S . We want to find a formula for computing the probability that the event A occurs before the event B in a sequence trials. Let $P(A)$ and $P(B)$ be the probabilities that A and B occur, respectively. Then the probability that neither A nor B occurs is $1 - P(A) - P(B)$. Let us denote this probability by r , that is $r = 1 - P(A) - P(B)$.

In the first trial, either A occurs, or B occurs, or neither A nor B occurs. In the first trial if A occurs, then the probability of A occurs before B is 1.

Example 2.4. A box of fuses contains 20 fuses, of which 5 are defective. If 3 of the fuses are selected at random and removed from the box in succession without replacement, what is the probability that all three fuses are defective?

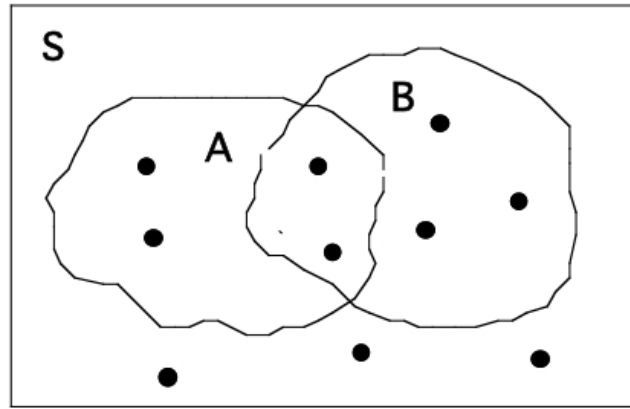
Answer: Let A be the event that the first fuse selected is defective. Let B be the event that the second fuse selected is defective. Let C be the event that the third fuse selected is defective. The probability that all three fuses selected are defective is $P(A \cap B \cap C)$. Hence

$$\begin{aligned} P(A \cap B \cap C) &= P(A) P(B/A) P(C/A \cap B) \\ &= \left(\frac{5}{20}\right) \left(\frac{4}{19}\right) \left(\frac{3}{18}\right) \\ &= \frac{1}{114}. \end{aligned}$$

Definition 2.3. Two events A and B of a sample space S are called independent if and only if

$$P(A \cap B) = P(A) P(B).$$

Example 2.5. The following diagram shows two events A and B in the sample space S . Are the events A and B independent?



Answer: There are 10 black dots in S and event A contains 4 of these dots. So the probability of A , is $P(A) = \frac{4}{10}$. Similarly, event B contains 5 black dots. Hence $P(B) = \frac{5}{10}$. The conditional probability of A given B is

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{2}{5}.$$

This shows that $P(A/B) = P(A)$. Hence A and B are independent.

Theorem 2.1. Let $A, B \subseteq S$. If A and B are independent and $P(B) > 0$, then

$$P(A/B) = P(A).$$

Proof:

$$\begin{aligned} P(A/B) &= \frac{P(A \cap B)}{P(B)} \\ &= \frac{P(A)P(B)}{P(B)} \\ &= P(A). \end{aligned}$$

Theorem 2.2. If A and B are independent events. Then A^c and B are independent. Similarly A and B^c are independent.

Proof: We know that A and B are independent, that is

$$P(A \cap B) = P(A)P(B)$$

and we want to show that A^c and B are independent, that is

$$P(A^c \cap B) = P(A^c)P(B).$$

Since

$$\begin{aligned} P(A^c \cap B) &= P(A^c/B)P(B) \\ &= [1 - P(A/B)]P(B) \\ &= P(B) - P(A/B)P(B) \\ &= P(B) - P(A \cap B) \\ &= P(B) - P(A)P(B) \\ &= P(B)[1 - P(A)] \\ &= P(B)P(A^c), \end{aligned}$$

the events A^c and B are independent. Similarly, it can be shown that A and B^c are independent and the proof is now complete.

Remark 2.1. The concept of independence is fundamental. In fact, it is this concept that justifies the mathematical development of probability as a separate discipline from measure theory. Mark Kac said, “independence of events is not a purely mathematical concept.” It can, however, be made plausible

that it should be interpreted by the rule of multiplication of probabilities and this leads to the mathematical definition of independence.

Example 2.6. Flip a coin and then independently cast a die. What is the probability of observing heads on the coin and a 2 or 3 on the die?

Answer: Let A denote the event of observing a head on the coin and let B be the event of observing a 2 or 3 on the die. Then

$$\begin{aligned} P(A \cap B) &= P(A) P(B) \\ &= \left(\frac{1}{2}\right) \left(\frac{2}{6}\right) \\ &= \frac{1}{6}. \end{aligned}$$

Example 2.7. An urn contains 3 red, 2 white and 4 yellow balls. An ordered sample of size 3 is drawn from the urn. If the balls are drawn with replacement so that one outcome does not change the probabilities of others, then what is the probability of drawing a sample that has balls of each color? Also, find the probability of drawing a sample that has two yellow balls and a red ball or a red ball and two white balls?

Answer:

$$P(RWY) = \left(\frac{3}{9}\right) \left(\frac{2}{9}\right) \left(\frac{4}{9}\right) = \frac{8}{243}$$

and

$$P(YYR \text{ or } RWW) = \left(\frac{4}{9}\right) \left(\frac{4}{9}\right) \left(\frac{3}{9}\right) + \left(\frac{3}{9}\right) \left(\frac{2}{9}\right) \left(\frac{2}{9}\right) = \frac{20}{243}.$$

If the balls are drawn without replacement, then

$$P(RWY) = \left(\frac{3}{9}\right) \left(\frac{2}{8}\right) \left(\frac{4}{7}\right) = \frac{1}{21}.$$

$$P(YYR \text{ or } RWW) = \left(\frac{4}{9}\right) \left(\frac{3}{8}\right) \left(\frac{3}{7}\right) + \left(\frac{3}{9}\right) \left(\frac{2}{8}\right) \left(\frac{1}{7}\right) = \frac{7}{84}.$$

There is a tendency to equate the concepts “mutually exclusive” and “independence”. This is a fallacy. Two events A and B are *mutually exclusive* if $A \cap B = \emptyset$ and they are called *possible* if $P(A) \neq 0 \neq P(B)$.

Theorem 2.2. Two possible mutually exclusive events are always dependent (that is not independent).

Proof: Suppose not. Then

$$P(A \cap B) = P(A) P(B)$$

$$P(\emptyset) = P(A) P(B)$$

$$0 = P(A) P(B).$$

Hence, we get either $P(A) = 0$ or $P(B) = 0$. This is a contradiction to the fact that A and B are possible events. This completes the proof.

Theorem 2.3. Two possible independent events are not mutually exclusive.

Proof: Let A and B be two independent events and suppose A and B are mutually exclusive. Then

$$P(A) P(B) = P(A \cap B)$$

$$= P(\emptyset)$$

$$= 0.$$

Therefore, we get either $P(A) = 0$ or $P(B) = 0$. This is a contradiction to the fact that A and B are possible events.

The possible events A and B exclusive implies A and B are not independent; and A and B independent implies A and B are not exclusive.

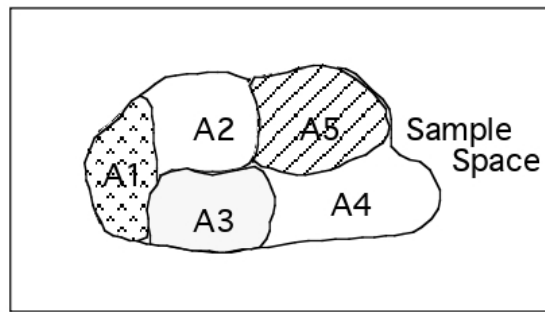
2.2. Bayes' Theorem

There are many situations where the ultimate outcome of an experiment depends on what happens in various intermediate stages. This issue is resolved by the Bayes' Theorem.

Definition 2.4. Let S be a set and let $\mathcal{P} = \{A_i\}_{i=1}^m$ be a collection of subsets of S . The collection $\mathcal{P}$ is called a partition of S if

$$(a) \quad S = \bigcup_{i=1}^m A_i$$

$$(b) \quad A_i \cap A_j = \emptyset \quad \text{for } i \neq j.$$



Theorem 2.4. If the events $\{B_i\}_{i=1}^m$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i = 1, 2, \dots, m$, then for any event A in S

$$P(A) = \sum_{i=1}^m P(B_i) P(A/B_i).$$

Proof: Let S be a sample space and A be an event in S . Let $\{B_i\}_{i=1}^m$ be any partition of S . Then

$$A = \bigcup_{i=1}^m (A \cap B_i).$$

Thus

$$\begin{aligned} P(A) &= \sum_{i=1}^m P(A \cap B_i) \\ &= \sum_{i=1}^m P(B_i) P(A/B_i). \end{aligned}$$

Theorem 2.5. If the events $\{B_i\}_{i=1}^m$ constitute a partition of the sample space S and $P(B_i) \neq 0$ for $i = 1, 2, \dots, m$, then for any event A in S such that $P(A) \neq 0$

$$P(B_k/A) = \frac{P(B_k) P(A/B_k)}{\sum_{i=1}^m P(B_i) P(A/B_i)} \quad k = 1, 2, \dots, m.$$

Proof: Using the definition of conditional probability, we get

$$P(B_k/A) = \frac{P(A \cap B_k)}{P(A)}.$$

Using Theorem 1, we get

$$P(B_k/A) = \frac{P(A \cap B_k)}{\sum_{i=1}^m P(B_i) P(A/B_i)}.$$

This completes the proof.

This Theorem is called Bayes Theorem. The probability $P(B_k)$ is called prior probability. The probability $P(B_k/A)$ is called posterior probability.

Example 2.8. Two boxes containing marbles are placed on a table. The boxes are labeled B_1 and B_2 . Box B_1 contains 7 green marbles and 4 white

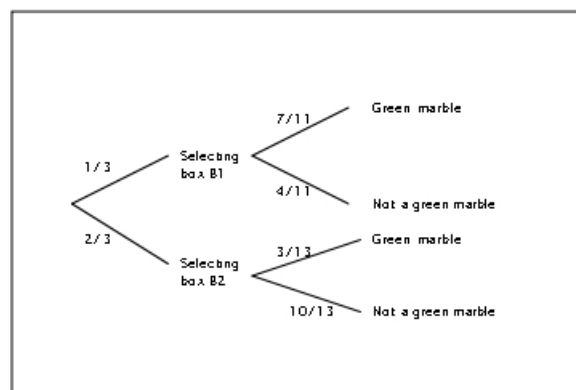
marbles. Box B_2 contains 3 green marbles and 10 yellow marbles. The boxes are arranged so that the probability of selecting box B_1 is $\frac{1}{3}$ and the probability of selecting box B_2 is $\frac{2}{3}$. Kathy is blindfolded and asked to select a marble. She will win a color TV if she selects a green marble. (a) What is the probability that Kathy will win the TV (that is, she will select a green marble)? (b) If Kathy wins the color TV, what is the probability that the green marble was selected from the first box?

Answer: Let A be the event of drawing a green marble. The prior probabilities are $P(B_1) = \frac{1}{3}$ and $P(B_2) = \frac{2}{3}$.

(a) The probability that Kathy will win the TV is

$$\begin{aligned}
 P(A) &= P(A \cap B_1) + P(A \cap B_2) \\
 &= P(A/B_1) P(B_1) + P(A/B_2) P(B_2) \\
 &= \left(\frac{7}{11}\right) \left(\frac{1}{3}\right) + \left(\frac{3}{13}\right) \left(\frac{2}{3}\right) \\
 &= \frac{7}{33} + \frac{2}{13} \\
 &= \frac{91}{429} + \frac{66}{429} \\
 &= \frac{157}{429}.
 \end{aligned}$$

(b) Given that Kathy won the TV, the probability that the green marble was selected from B_1 is

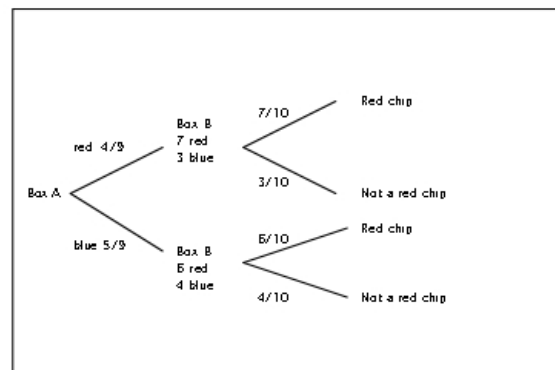


$$\begin{aligned}
 P(B_1/A) &= \frac{P(A/B_1) P(B_1)}{P(A/B_1) P(B_1) + P(A/B_2) P(B_2)} \\
 &= \frac{\left(\frac{7}{11}\right) \left(\frac{1}{3}\right)}{\left(\frac{7}{11}\right) \left(\frac{1}{3}\right) + \left(\frac{3}{13}\right) \left(\frac{2}{3}\right)} \\
 &= \frac{91}{157}.
 \end{aligned}$$

Note that $P(A/B_1)$ is the probability of selecting a green marble from B_1 whereas $P(B_1/A)$ is the probability that the green marble was selected from box B_1 .

Example 2.9. Suppose box A contains 4 red and 5 blue chips and box B contains 6 red and 3 blue chips. A chip is chosen at random from the box A and placed in box B . Finally, a chip is chosen at random from among those now in box B . What is the probability a blue chip was transferred from box A to box B given that the chip chosen from box B is red?

Answer: Let E represent the event of moving a blue chip from box A to box B . We want to find the probability of a blue chip which was moved from box A to box B given that the chip chosen from B was red. The probability of choosing a red chip from box A is $P(R) = \frac{4}{9}$ and the probability of choosing a blue chip from box A is $P(B) = \frac{5}{9}$. If a red chip was moved from box A to box B , then box B has 7 red chips and 3 blue chips. Thus the probability of choosing a red chip from box B is $\frac{7}{10}$. Similarly, if a blue chip was moved from box A to box B , then the probability of choosing a red chip from box B is $\frac{6}{10}$.



Hence, the probability that a blue chip was transferred from box A to box B given that the chip chosen from box B is red is given by

$$\begin{aligned} P(E/R) &= \frac{P(R/E) P(E)}{P(R)} \\ &= \frac{\left(\frac{6}{10}\right) \left(\frac{5}{9}\right)}{\left(\frac{7}{10}\right) \left(\frac{4}{9}\right) + \left(\frac{6}{10}\right) \left(\frac{5}{9}\right)} \\ &= \frac{15}{29}. \end{aligned}$$

Example 2.10. Sixty percent of new drivers have had driver education. During their first year, new drivers without driver education have probability 0.08 of having an accident, but new drivers with driver education have only a 0.05 probability of an accident. What is the probability a new driver has had driver education, given that the driver has had no accident the first year?

Answer: Let A represent the new driver who has had driver education and B represent the new driver who has had an accident in his first year. Let A^c and B^c be the complement of A and B , respectively. We want to find the probability that a new driver has had driver education, given that the driver has had no accidents in the first year, that is $P(A/B^c)$.

$$\begin{aligned} P(A/B^c) &= \frac{P(A \cap B^c)}{P(B^c)} \\ &= \frac{P(B^c/A) P(A)}{P(B^c/A) P(A) + P(B^c/A^c) P(A^c)} \\ &= \frac{[1 - P(B/A)] P(A)}{[1 - P(B/A)] P(A) + [1 - P(B/A^c)] [1 - P(A)]} \\ &= \frac{\left(\frac{60}{100}\right) \left(\frac{95}{100}\right)}{\left(\frac{40}{100}\right) \left(\frac{92}{100}\right) + \left(\frac{60}{100}\right) \left(\frac{95}{100}\right)} \\ &= 0.6077. \end{aligned}$$

Example 2.11. One-half percent of the population has AIDS. There is a test to detect AIDS. A positive test result is supposed to mean that you

have AIDS but the test is not perfect. For people with AIDS, the test misses the diagnosis 2% of the times. And for the people without AIDS, the test incorrectly tells 3% of them that they have AIDS. (a) What is the probability that a person picked at random will test positive? (b) What is the probability that you have AIDS given that your test comes back positive?

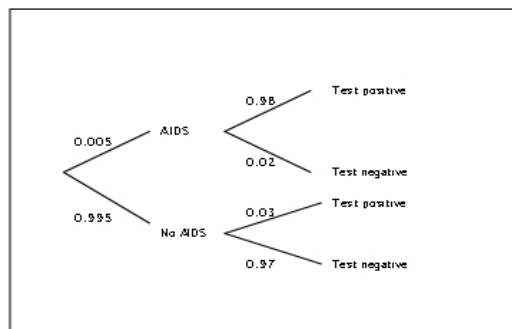
Answer: Let A denote the event of one who has AIDS and let B denote the event that the test comes out positive.

(a) The probability that a person picked at random will test positive is given by

$$\begin{aligned} P(\text{test positive}) &= (0.005)(0.98) + (0.995)(0.03) \\ &= 0.0049 + 0.0298 = 0.035. \end{aligned}$$

(b) The probability that you have AIDS given that your test comes back positive is given by

$$\begin{aligned} P(A/B) &= \frac{\text{favorable positive branches}}{\text{total positive branches}} \\ &= \frac{(0.005)(0.98)}{(0.005)(0.98) + (0.995)(0.03)} \\ &= \frac{0.0049}{0.035} = 0.14. \end{aligned}$$



Remark 2.2. This example illustrates why Bayes' theorem is so important. What we would really like to know in this situation is a first-stage result: Do you have AIDS? But we cannot get this information without an autopsy. The first stage is hidden. But the second stage is not hidden. The best we can do is make a prediction about the first stage. This illustrates why backward conditional probabilities are so useful.

RANDOM VARIABLES AND DISTRIBUTION FUNCTIONS

3.1. Introduction

In many random experiments, the elements of sample space are not necessarily numbers. For example, in a coin tossing experiment the sample space consists of

$$S = \{\text{Head}, \text{Tail}\}.$$

Statistical methods involve primarily numerical data. Hence, one has to ‘mathematize’ the outcomes of the sample space. This mathematization, or quantification, is achieved through the notion of random variables.

Definition 3.1. Consider a random experiment whose sample space is S . A *random variable* X is a function from the sample space S into the set of real numbers $\mathbb{R}$ such that for each interval I in $\mathbb{R}$, the set $\{s \in S \mid X(s) \in I\}$ is an event in S .

In a particular experiment a random variable X would be some function that assigns a real number $X(s)$ to each possible outcome s in the sample space. Given a random experiment, there can be many random variables. This is due to the fact that given two (finite) sets A and B , the number of distinct functions one can come up with is $|B|^{|A|}$. Here $|A|$ means the cardinality of the set A .

Random variable is not a variable. Also, it is not random. Thus someone named it inappropriately. The following analogy speaks the role of the random variable. Random variable is like the *Holy Roman Empire* – it was

not holy, it was not Roman, and it was not an empire. A random variable is neither random nor variable, it is simply a function. The values it takes on are both random and variable.

Definition 3.2. The set $\{x \in \mathbb{R} \mid x = X(s), s \in S\}$ is called the *space of the random variable* X .

The space of the random variable X will be denoted by R_X . The space of the random variable X is actually the range of the function $X : S \rightarrow \mathbb{R}$.

Example 3.1. Consider the coin tossing experiment. Construct a random variable X for this experiment. What is the space of this random variable X ?

Answer: The sample space of this experiment is given by

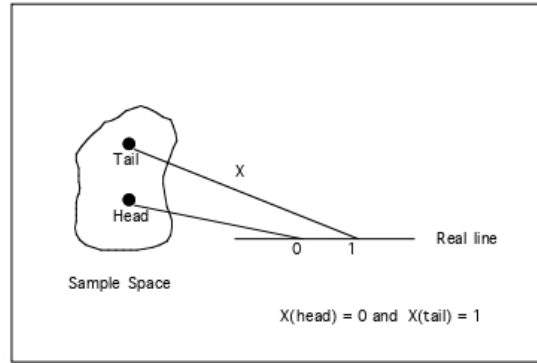
$$S = \{\text{Head}, \text{Tail}\}.$$

Let us define a function from S into the set of reals as follows

$$\begin{aligned} X(\text{Head}) &= 0 \\ X(\text{Tail}) &= 1. \end{aligned}$$

Then X is a valid map and thus by our definition of random variable, it is a random variable for the coin tossing experiment. The space of this random variable is

$$R_X = \{0, 1\}.$$



Example 3.2. Consider an experiment in which a coin is tossed ten times. What is the sample space of this experiment? How many elements are in this sample space? Define a random variable for this sample space and then find the space of the random variable.

Answer: The sample space of this experiment is given by

$$S = \{s \mid s \text{ is a sequence of 10 heads or tails}\}.$$

The cardinality of S is

$$|S| = 2^{10}.$$

Let $X : S \rightarrow \mathbb{R}$ be a function from the sample space S into the set of reals $\mathbb{R}$ defined as follows:

$$X(s) = \text{number of heads in sequence } s.$$

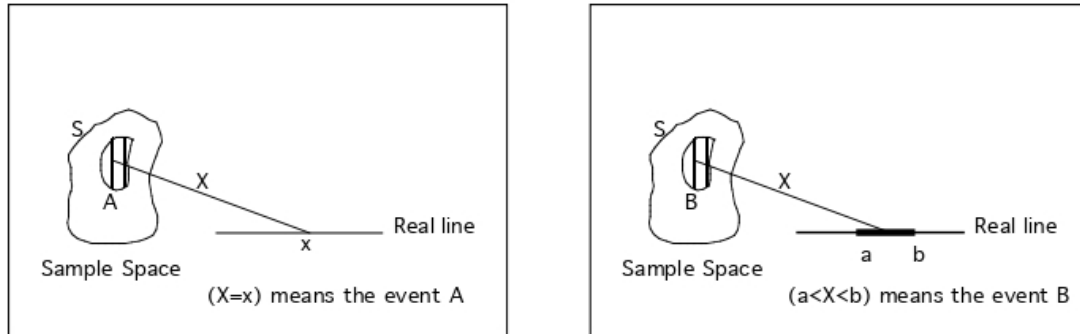
Then X is a random variable. This random variable, for example, maps the sequence $HHTTTHTTTHH$ to the real number 5, that is

$$X(HHTTTHTTTHH) = 5.$$

The space of this random variable is

$$R_X = \{0, 1, 2, \dots, 10\}.$$

Now, we introduce some notations. By $(X = x)$ we mean the event $\{s \in S \mid X(s) = x\}$. Similarly, $(a < X < b)$ means the event $\{s \in S \mid a < X < b\}$ of the sample space S . These are illustrated in the following diagrams.



There are three types of random variables: discrete, continuous, and mixed. However, in most applications we encounter either discrete or continuous random variable. In this book we only treat these two types of random variables. First, we consider the discrete case and then we examine the continuous case.

Definition 3.3. If the space of random variable X is countable, then X is called a *discrete random variable*.

Distribution Functions of Discrete Random Variables

Every random variable is characterized through its probability density function.

Definition Let R_X be the space of the random variable X . The function $f : R_X \rightarrow \mathbb{R}$ defined by

$$f(x) = P(X = x)$$

is called the *probability density function* (pdf) of X .

Example In an introductory statistics class of 50 students, there are 11 freshman, 19 sophomores, 14 juniors and 6 seniors. One student is selected at random. What is the sample space of this experiment? Construct a random variable X for this sample space and then find its space. Further, find the probability density function of this random variable X .

Answer: The sample space of this random experiment is

$$S = \{Fr, So, Jr, Sr\}.$$

Define a function $X : S \rightarrow \mathbb{R}$ as follows:

$$\begin{aligned} X(Fr) &= 1, & X(So) &= 2 \\ X(Jr) &= 3, & X(Sr) &= 4. \end{aligned}$$

Then clearly X is a random variable in S . The space of X is given by

$$R_X = \{1, 2, 3, 4\}.$$

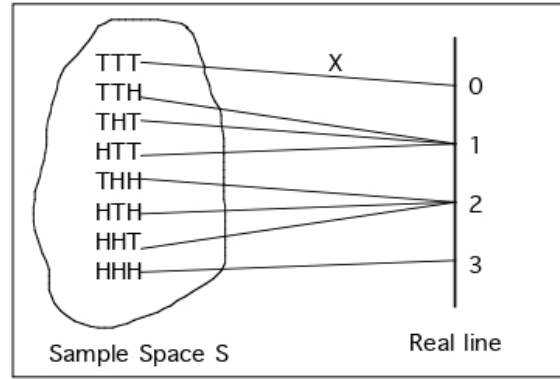
The probability density function of X is given by

$$\begin{aligned} f(1) &= P(X = 1) = \frac{11}{50} \\ f(2) &= P(X = 2) = \frac{19}{50} \\ f(3) &= P(X = 3) = \frac{14}{50} \\ f(4) &= P(X = 4) = \frac{6}{50}. \end{aligned}$$

Example A fair coin is tossed 3 times. Let the random variable X denote the number of heads in 3 tosses of the coin. Find the sample space, the space of the random variable, and the probability density function of X .

Answer: The sample space S of this experiment consists of all binary sequences of length 3, that is

$$S = \{TTT, TTH, THT, HTT, THH, HTH, HHT, HHH\}.$$



The space of this random variable is given by

$$R_X = \{0, 1, 2, 3\}.$$

Therefore, the probability density function of X is given by

$$\begin{aligned} f(0) &= P(X = 0) = \frac{1}{8} \\ f(1) &= P(X = 1) = \frac{3}{8} \\ f(2) &= P(X = 2) = \frac{3}{8} \\ f(3) &= P(X = 3) = \frac{1}{8}. \end{aligned}$$

This can be written as follows:

$$f(x) = \binom{3}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{3-x} \quad x = 0, 1, 2, 3.$$

Theorem . If X is a discrete random variable with space R_X and probability density function $f(x)$, then

(a) $f(x) \geq 0$ for all x in R_X , and

(b) $\sum_{x \in R_X} f(x) = 1$.

Definition . The cumulative distribution function $F(x)$ of a random variable X is defined as

$$F(x) = P(X \leq x)$$

for all real numbers x .

Theorem . If X is a random variable with the space R_X , then

$$F(x) = \sum_{t \leq x} f(t)$$

for $x \in R_X$.

$F(x)$ represents the accumulation of $f(t)$ up to $t \leq x$.

Theorem . Let X be a random variable with cumulative distribution function $F(x)$. Then the cumulative distribution function satisfies the followings:

(a) $F(-\infty) = 0$,

(b) $F(\infty) = 1$, and

(c) $F(x)$ is an increasing function, that is if $x < y$, then $F(x) \leq F(y)$ for all reals x, y .

The proof of this theorem is trivial and we leave it to the students.

Theorem . If the space R_X of the random variable X is given by $R_X = \{x_1 < x_2 < x_3 < \dots < x_n\}$, then

$$f(x_1) = F(x_1)$$

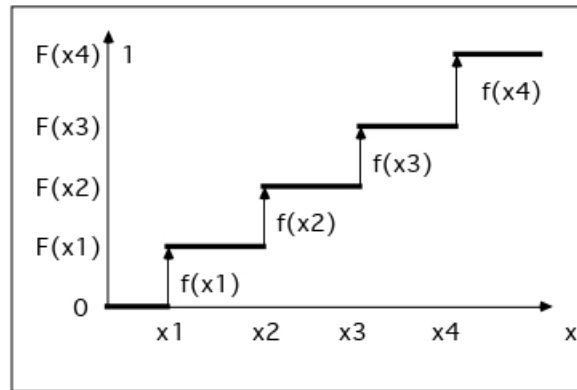
$$f(x_2) = F(x_2) - F(x_1)$$

$$f(x_3) = F(x_3) - F(x_2)$$

..

..

$$f(x_n) = F(x_n) - F(x_{n-1}).$$



Distribution Functions of Continuous Random Variables

A random variable X is said to be continuous if its space is either an interval or a union of intervals. The following definition formally defines a continuous random variable.

Definition A random variable X is said to be a continuous random variable if there exists a continuous function $f : \mathbb{R} \rightarrow [0, \infty)$ such that for every set of real numbers A

$$P(X \in A) = \int_A f(x) dx.$$

Definition The function f is called the probability density function of the continuous random variable X .

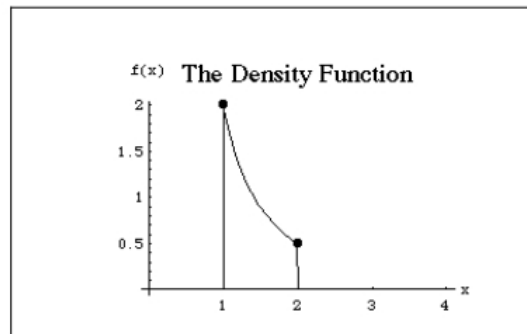
It can be easily shown that for every probability density function f ,

$$\int_{-\infty}^{\infty} f(x) dx = 1.$$

Example Is the real valued function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 2x^{-2} & \text{if } 1 < x < 2 \\ 0 & \text{otherwise,} \end{cases}$$

a probability density function for some random variable X ?



Answer: We have to show that f is nonnegative and the area under $f(x)$ is unity. Since the domain of f is the interval $(0, 1)$, it is clear that f is nonnegative. Next, we calculate

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_1^2 2x^{-2} dx \\ &= -2 \left[\frac{1}{x} \right]_1^2 \\ &= -2 \left[\frac{1}{2} - 1 \right] \\ &= 1. \end{aligned}$$

Thus f is a probability density function.

Percentiles for Continuous Random Variables

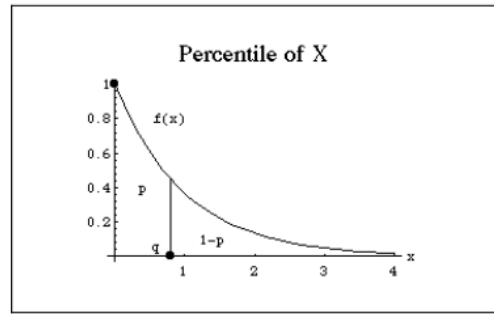
In this section, we discuss various percentiles of a continuous random variable. If the random variable is discrete, then to discuss percentile, we have to know the order statistics of samples. We shall treat the percentile of discrete random variable in Chapter 13.

Definition Let p be a real number between 0 and 1. A $100p^{\text{th}}$ percentile of the distribution of a random variable X is any real number q satisfying

$$P(X \leq q) \leq p \quad \text{and} \quad P(X > q) \leq 1 - p.$$

A $100p^{\text{th}}$ percentile is a measure of location for the probability distribution in the sense that q divides the distribution of the probability mass into

two parts, one having probability mass p and other having probability mass $1 - p$ (see diagram below).



Example If the random variable X has the density function

$$f(x) = \begin{cases} e^{x-2} & \text{for } x < 2 \\ 0 & \text{otherwise,} \end{cases}$$

then what is the 75th percentile of X ?

Answer: Since $100p^{\text{th}} = 75$, we get $p = 0.75$. By definition of percentile, we have

$$\begin{aligned} 0.75 = p &= \int_{-\infty}^q f(x) dx \\ &= \int_{-\infty}^q e^{x-2} dx \\ &= [e^{x-2}]_{-\infty}^q \\ &= e^{q-2}. \end{aligned}$$

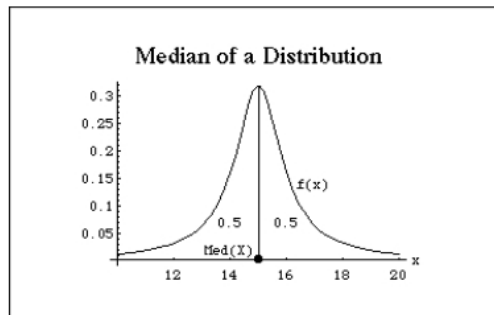
From this solving for q , we get the 75th percentile to be

$$q = 2 + \ln\left(\frac{3}{4}\right).$$

Definition The 25th and 75th percentiles of any distribution are called the *first* and the *third* quartiles, respectively.

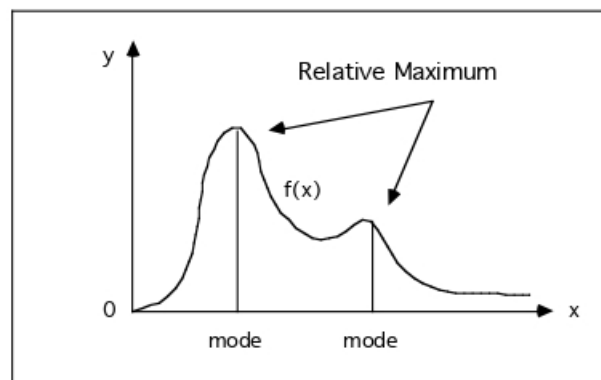
Definition The 50th percentile of any distribution is called the *median* of the distribution.

The median divides the distribution of the probability mass into two equal parts (see the following figure).



If a probability density function $f(x)$ is symmetric about the y -axis, then the median is always 0.

Definition A mode of the distribution of a continuous random variable X is the value of x where the probability density function $f(x)$ attains a relative maximum (see diagram).



A mode of a random variable X is one of its most probable values. A random variable can have more than one mode.

Moments of Random Variables

In this chapter, we introduce the concepts of various moments of a random variable. Further, we examine the expected value and the variance of random variables in detail. We shall conclude this chapter with a discussion of Chebychev's inequality.

Definition . The n^{th} moment about the origin of a random variable X , as denoted by $E(X^n)$, is defined to be

$$E(X^n) = \begin{cases} \sum_{x \in R_X} x^n f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^n f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

for $n = 0, 1, 2, 3, \dots$, provided the right side converges absolutely.

If $n = 1$, then $E(X)$ is called the first moment about the origin. If $n = 2$, then $E(X^2)$ is called the second moment of X about the origin. In general, these moments may or may not exist for a given random variable. If for a random variable, a particular moment does not exist, then we say that the random variable does not have that moment. For these moments to exist one requires absolute convergence of the sum or the integral. Next, we shall define two important characteristics of a random variable, namely the expected value and variance. Occasionally $E(X^n)$ will be written as $E[X^n]$.

Expected Value of Random Variables

A random variable X is characterized by its probability density function, which defines the relative likelihood of assuming one value over the others.

Definition . Let X be a random variable with space R_X and probability density function $f(x)$. The mean μ_X of the random variable X is defined as

$$\mu_X = \begin{cases} \sum_{x \in R_X} x f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

if the right hand side exists.

The mean of a random variable is a composite of its values weighted by the corresponding probabilities. The mean is a measure of central tendency: the value that the random variable takes "on average." The mean is also called the expected value of the random variable X and is denoted by $E(X)$. The symbol E is called the expectation operator. The expected value of a random variable may or may not exist.

Theorem Let X be a random variable with pdf $f(x)$. If a and b are any two real numbers, then

$$E(aX + b) = a E(X) + b.$$

Proof: We will prove only for the continuous case.

$$\begin{aligned} E(aX + b) &= \int_{-\infty}^{\infty} (ax + b) f(x) dx \\ &= \int_{-\infty}^{\infty} ax f(x) dx + \int_{-\infty}^{\infty} b f(x) dx \\ &= a \int_{-\infty}^{\infty} x f(x) dx + b \\ &= aE(X) + b. \end{aligned}$$

To prove the discrete case, replace the integral by summation. This completes the proof.

Variance of Random Variables

The spread of the distribution of a random variable X is its variance.

Definition Let X be a random variable with mean μ_X . The variance of X , denoted by $Var(X)$, is defined as

$$Var(X) = E([X - \mu_X]^2).$$

It is also denoted by σ_X^2 . The positive square root of the variance is called the standard deviation of the random variable X . Like variance, the standard deviation also measures the spread. The following theorem tells us how to compute the variance in an alternative way.

Theorem If X is a random variable with mean μ_X and variance σ_X^2 , then

$$\sigma_X^2 = E(X^2) - (\mu_X)^2.$$

Proof:

$$\begin{aligned}
\sigma_X^2 &= E([X - \mu_X]^2) \\
&= E(X^2 - 2\mu_X X + \mu_X^2) \\
&= E(X^2) - 2\mu_X E(X) + (\mu_X)^2 \\
&= E(X^2) - 2\mu_X \mu_X + (\mu_X)^2 \\
&= E(X^2) - (\mu_X)^2.
\end{aligned}$$

Theorem If X is a random variable with mean μ_X and variance σ_X^2 , then

$$\text{Var}(aX + b) = a^2 \text{Var}(X),$$

where a and b are arbitrary real constants.

Proof:

$$\begin{aligned}
\text{Var}(aX + b) &= E([(aX + b) - \mu_{aX+b}]^2) \\
&= E([aX + b - E(aX + b)]^2) \\
&= E([aX + b - a\mu_X - b]^2) \\
&= E(a^2 [X - \mu_X]^2) \\
&= a^2 E([X - \mu_X]^2) \\
&= a^2 \text{Var}(X).
\end{aligned}$$

Moment Generating Functions

Definition Let X be a random variable whose probability density function is $f(x)$. A real valued function $M : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$M(t) = E(e^{tX})$$

is called the moment generating function of X if this expected value exists for all t in the interval $-h < t < h$ for some $h > 0$.

In general, not every random variable has a moment generating function. But if the moment generating function of a random variable exists, then it is unique. At the end of this section, we will give an example of a random variable which does not have a moment generating function.

Using the definition of expected value of a random variable, we obtain the explicit representation for $M(t)$ as

$$M(t) = \begin{cases} \sum_{x \in R_X} e^{tx} f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} e^{tx} f(x) dx & \text{if } X \text{ is continuous.} \end{cases}$$

Example . Let X be a random variable whose moment generating function is $M(t)$ and n be any natural number. What is the n^{th} derivative of $M(t)$ at $t = 0$?

Answer:

$$\begin{aligned} \frac{d}{dt} M(t) &= \frac{d}{dt} E(e^{tX}) \\ &= E\left(\frac{d}{dt} e^{tX}\right) \\ &= E(X e^{tX}). \end{aligned}$$

Similarly,

$$\begin{aligned} \frac{d^2}{dt^2} M(t) &= \frac{d^2}{dt^2} E(e^{tX}) \\ &= E\left(\frac{d^2}{dt^2} e^{tX}\right) \\ &= E(X^2 e^{tX}). \end{aligned}$$

Hence, in general we get

$$\begin{aligned}\frac{d^n}{dt^n}M(t) &= \frac{d^n}{dt^n}E(e^{tX}) \\ &= E\left(\frac{d^n}{dt^n}e^{tX}\right) \\ &= E(X^n e^{tX}).\end{aligned}$$

If we set $t = 0$ in the n^{th} derivative, we get

$$\left.\frac{d^n}{dt^n}M(t)\right|_{t=0} = E(X^n e^{tX})|_{t=0} = E(X^n).$$

Hence the n^{th} derivative of the moment generating function of X evaluated at $t = 0$ is the n^{th} moment of X about the origin.

This example tells us if we know the moment generating function of a random variable; then we can generate all the moments of X by taking derivatives of the moment generating function and then evaluating them at zero.

Example What is the moment generating function of the random variable X whose probability density function is given by

$$f(x) = \begin{cases} e^{-x} & \text{for } x > 0 \\ 0 & \text{otherwise?} \end{cases}$$

What are the mean and variance of X ?

Answer: The moment generating function of X is

$$\begin{aligned}M(t) &= E(e^{tX}) \\ &= \int_0^\infty e^{tx} f(x) dx \\ &= \int_0^\infty e^{tx} e^{-x} dx \\ &= \int_0^\infty e^{-(1-t)x} dx \\ &= \frac{1}{1-t} \left[-e^{-(1-t)x} \right]_0^\infty \\ &= \frac{1}{1-t} \quad \text{if } 1-t > 0.\end{aligned}$$

The expected value of X can be computed from $M(t)$ as

$$\begin{aligned} E(X) &= \left. \frac{d}{dt} M(t) \right|_{t=0} \\ &= \left. \frac{d}{dt} (1-t)^{-1} \right|_{t=0} \\ &= (1-t)^{-2} \Big|_{t=0} \\ &= \left. \frac{1}{(1-t)^2} \right|_{t=0} \\ &= 1. \end{aligned}$$

Similarly

$$\begin{aligned} E(X^2) &= \left. \frac{d^2}{dt^2} M(t) \right|_{t=0} \\ &= \left. \frac{d^2}{dt^2} (1-t)^{-1} \right|_{t=0} \\ &= 2(1-t)^{-3} \Big|_{t=0} \\ &= \left. \frac{2}{(1-t)^3} \right|_{t=0} \\ &= 2. \end{aligned}$$

Therefore, the variance of X is

$$Var(X) = E(X^2) - (\mu)^2 = 2 - 1 = 1.$$

Example Let X have the probability density function

$$f(x) = \begin{cases} \frac{1}{9} \left(\frac{8}{9}\right)^x & \text{for } x = 0, 1, 2, \dots, \infty \\ 0 & \text{otherwise.} \end{cases}$$

What is the moment generating function of the random variable X ?

Answer:

$$\begin{aligned} M(t) &= E(e^{tX}) \\ &= \sum_{x=0}^{\infty} e^{tx} f(x) \\ &= \sum_{x=0}^{\infty} e^{tx} \left(\frac{1}{9}\right) \left(\frac{8}{9}\right)^x \\ &= \left(\frac{1}{9}\right) \sum_{x=0}^{\infty} \left(e^t \frac{8}{9}\right)^x \\ &= \left(\frac{1}{9}\right) \frac{1}{1 - e^t \frac{8}{9}} \quad \text{if } e^t \frac{8}{9} < 1 \\ &= \frac{1}{9 - 8e^t} \quad \text{if } t < \ln\left(\frac{9}{8}\right). \end{aligned}$$

Example Let the random variable X have moment generating function $M(t) = (1 - t)^{-2}$ for $t < 1$. What is the third moment of X about the origin?

Answer: To compute the third moment $E(X^3)$ of X about the origin, we

need to compute the third derivative of $M(t)$ at $t = 0$.

$$\begin{aligned} M(t) &= (1 - t)^{-2} \\ M'(t) &= 2 (1 - t)^{-3} \\ M''(t) &= 6 (1 - t)^{-4} \\ M'''(t) &= 24 (1 - t)^{-5}. \end{aligned}$$

Thus the third moment of X is given by

$$E(X^3) = \frac{24}{(1 - 0)^5} = 24.$$

Theorem Let X be a random variable with the moment generating function $M_X(t)$. If a and b are any two real constants, then

$$\begin{aligned} M_{X+a}(t) &= e^{at} M_X(t) \\ M_{bX}(t) &= M_X(bt) \\ M_{\frac{X+a}{b}}(t) &= e^{\frac{a}{b}t} M_X\left(\frac{t}{b}\right). \end{aligned}$$

Bivariate Discrete Random Variables

In this section, we develop all the necessary terminologies for studying bivariate discrete random variables.

Definition A discrete *bivariate random variable* (X, Y) is an ordered pair of discrete random variables.

Definition Let (X, Y) be a bivariate random variable and let R_X and R_Y be the range spaces of X and Y , respectively. A real-valued function $f : R_X \times R_Y \rightarrow \mathbb{R}$ is called a *joint probability density function* for X and Y if and only if

$$f(x, y) = P(X = x, Y = y)$$

for all $(x, y) \in R_X \times R_Y$. Here, the event $(X = x, Y = y)$ means the intersection of the events $(X = x)$ and $(Y = y)$, that is

$$(X = x) \cap (Y = y).$$

A real valued function f of two variables is a joint probability density function of a pair of discrete random variables X and Y (with range spaces R_X and R_Y , respectively) if and only if

$$(a) \quad f(x, y) \geq 0 \quad \text{for all } (x, y) \in R_X \times R_Y;$$

$$(b) \quad \sum_{x \in R_X} \sum_{y \in R_Y} f(x, y) = 1.$$

Definition Let (X, Y) be a discrete bivariate random variable. Let R_X and R_Y be the range spaces of X and Y , respectively. Let $f(x, y)$ be the joint probability density function of X and Y . The function

$$f_1(x) = \sum_{y \in R_Y} f(x, y)$$

is called the *marginal probability density function* of X . Similarly, the function

$$f_2(y) = \sum_{x \in R_X} f(x, y)$$

is called the *marginal probability density function* of Y .

Definition Let X and Y be any two discrete random variables. The real valued function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ is called the *joint cumulative probability distribution function* of X and Y if and only if

$$F(x, y) = P(X \leq x, Y \leq y)$$

for all $(x, y) \in \mathbb{R}^2$. Here, the event $(X \leq x, Y \leq y)$ means $(X \leq x) \cap (Y \leq y)$.

From this definition it can be shown that for any real numbers a and b

$$F(a \leq X \leq b, c \leq Y \leq d) = F(b, d) + F(a, c) - F(a, d) - F(b, c).$$

Further, one can also show that

$$F(x, y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t)$$

where (s, t) is any pair of nonnegative numbers.

Bivariate Continuous Random Variables

In this section, we shall extend the idea of probability density functions of one random variable to that of two random variables.

Definition The *joint probability density function* of the random variables X and Y is an integrable function $f(x, y)$ such that

- (a) $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$; and
- (b) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$.

Example Let the joint density function of X and Y be given by

$$f(x, y) = \begin{cases} kxy^2 & \text{if } 0 < x < y < 1 \\ 0 & \text{otherwise.} \end{cases}$$

What is the value of the constant k ?

Answer: Since f is a joint probability density function, we have

$$\begin{aligned}
 1 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy \\
 &= \int_0^1 \int_0^y k x y^2 dx dy \\
 &= \int_0^1 k y^2 \int_0^y x dx dy \\
 &= \frac{k}{2} \int_0^1 y^4 dy \\
 &= \frac{k}{10} [y^5]_0^1 \\
 &= \frac{k}{10}.
 \end{aligned}$$

Hence $k = 10$.

If we know the joint probability density function f of the random variables X and Y , then we can compute the probability of the event A from

$$P(A) = \int \int_A f(x, y) dx dy.$$

Definition Let (X, Y) be a continuous bivariate random variable. Let $f(x, y)$ be the joint probability density function of X and Y . The function

$$f_1(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

is called the *marginal probability density function* of X . Similarly, the function

$$f_2(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

is called the *marginal probability density function* of Y .

Definition Let X and Y be the continuous random variables with joint probability density function $f(x, y)$. The *joint cumulative distribution function* $F(x, y)$ of X and Y is defined as

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(u, v) du dv$$

for all $(x, y) \in \mathbb{R}^2$.

From the fundamental theorem of calculus, we again obtain

$$f(x, y) = \frac{\partial^2 F}{\partial x \partial y}.$$

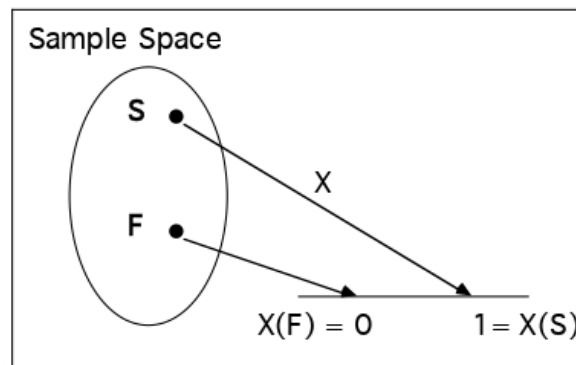
SOME SPECIAL DISCRETE DISTRIBUTIONS

Given a random experiment, we can find the set of all possible outcomes which is known as the sample space. Objects in a sample space may not be numbers. Thus, we use the notion of random variable to quantify the qualitative elements of the sample space. A random variable is characterized by either its probability density function or its cumulative distribution function. The other characteristics of a random variable are its mean, variance and moment generating function. In this chapter, we explore some frequently encountered discrete distributions and study their important characteristics.

Bernoulli Distribution

A Bernoulli trial is a random experiment in which there are precisely two possible outcomes, which we conveniently call ‘failure’ (F) and ‘success’ (S). We can define a random variable from the sample space $\{S, F\}$ into the set of real numbers as follows:

$$X(F) = 0 \qquad X(S) = 1.$$



The probability density function of this random variable is

$$\begin{aligned} f(0) &= P(X = 0) = 1 - p \\ f(1) &= P(X = 1) = p, \end{aligned}$$

where p denotes the probability of success. Hence

$$f(x) = p^x (1 - p)^{1-x}, \quad x = 0, 1.$$

Definition The random variable X is called the Bernoulli random variable if its probability density function is of the form

$$f(x) = p^x (1 - p)^{1-x}, \quad x = 0, 1$$

where p is the probability of success.

We denote the Bernoulli random variable by writing $X \sim BER(p)$.

Theorem If X is a Bernoulli random variable with parameter p , then the mean, variance and moment generating functions are respectively given by

$$\begin{aligned}\mu_X &= p \\ \sigma_X^2 &= p(1-p) \\ M_X(t) &= (1-p) + pe^t.\end{aligned}$$

Binomial Distribution

Consider a fixed number n of mutually independent Bernoulli trials. Suppose these trials have same probability of success, say p . A random variable X is called a binomial random variable if it represents the total number of successes in n independent Bernoulli trials.

Now we determine the probability density function of a binomial random variable. Recall that the probability density function of X is defined as

$$f(x) = P(X = x).$$

Thus, to find the probability density function of X we have to find the probability of x successes in n independent trials.

If we have x successes in n trials, then the probability of each n -tuple with x successes and $n - x$ failures is

$$p^x (1-p)^{n-x}.$$

However, there are $\binom{n}{x}$ tuples with x successes and $n - x$ failures in n trials. Hence

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Therefore, the probability density function of X is

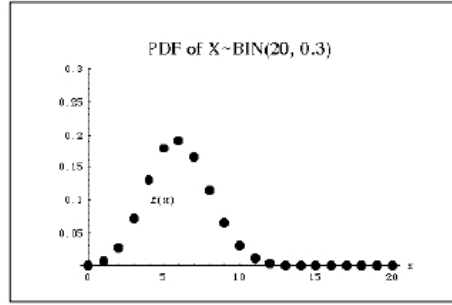
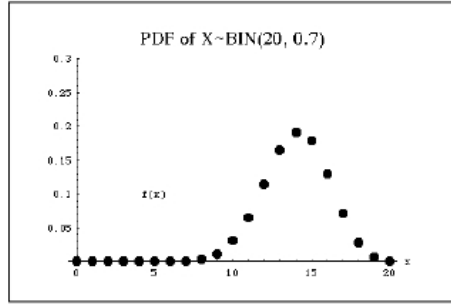
$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Definition The random variable X is called the binomial random variable with parameters p and n if its probability density function is of the form

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

where $0 < p < 1$ is the probability of success.

We will denote a binomial random variable with parameters p and n as $X \sim \text{BIN}(n, p)$.



Example Is the real valued function $f(x)$ given by

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

where n and p are parameters, a probability density function?

Answer: To answer this question, we have to check that $f(x)$ is nonnegative and $\sum_{x=0}^n f(x)$ is 1. It is easy to see that $f(x) \geq 0$. We show that sum is one.

$$\begin{aligned} \sum_{x=0}^n f(x) &= \sum_{x=0}^n \binom{n}{x} p^x (1-p)^{n-x} \\ &= (p + 1 - p)^n \\ &= 1. \end{aligned}$$

Hence $f(x)$ is really a probability density function.

Example On a five-question multiple-choice test there are five possible answers, of which one is correct. If a student guesses randomly and independently, what is the probability that she is correct only on questions 1 and 4?

Answer: Here the probability of success is $p = \frac{1}{5}$, and thus $1 - p = \frac{4}{5}$. Therefore, the probability that she is correct on questions 1 and 4 is

$$\begin{aligned} P(\text{correct on questions 1 and 4}) &= p^2 (1-p)^3 \\ &= \left(\frac{1}{5}\right)^2 \left(\frac{4}{5}\right)^3 \\ &= \frac{64}{5^5} = 0.02048. \end{aligned}$$

Theorem If X is binomial random variable with parameters p and n , then the mean, variance and moment generating functions are respectively given by

$$\begin{aligned}\mu_X &= n p \\ \sigma_X^2 &= n p (1 - p) \\ M_X(t) &= [(1 - p) + p e^t]^n.\end{aligned}$$

Proof: First, we determine the moment generating function $M(t)$ of X and then we generate mean and variance from $M(t)$.

$$\begin{aligned}M(t) &= E(e^{tX}) \\ &= \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1 - p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (p e^t)^x (1 - p)^{n-x} \\ &= (p e^t + 1 - p)^n.\end{aligned}$$

Hence

$$M'(t) = n (p e^t + 1 - p)^{n-1} p e^t.$$

Therefore

$$\mu_X = M'(0) = n p.$$

Similarly

$$M''(t) = n (p e^t + 1 - p)^{n-1} p e^t + n(n-1) (p e^t + 1 - p)^{n-2} (p e^t)^2.$$

Therefore

$$E(X^2) = M''(0) = n(n-1)p^2 + n p.$$

Hence

$$Var(X) = E(X^2) - \mu_X^2 = n(n-1)p^2 + n p - n^2 p^2 = n p (1 - p).$$

This completes the proof.

Geometric Distribution

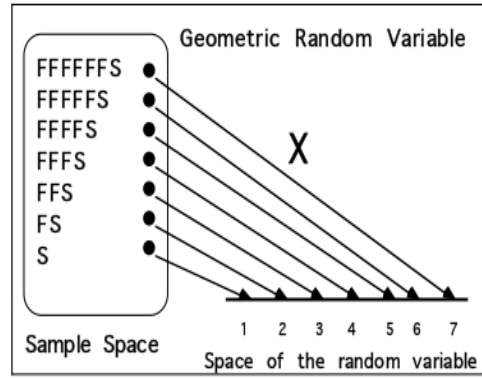
If X represents the total number of successes in n independent Bernoulli trials, then the random variable

$$X \sim BIN(n, p),$$

where p is the probability of success of a single Bernoulli trial and the probability density function of X is given by

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Let X denote the trial number on which the first success occurs.



Hence the probability that the first success occurs on x^{th} trial is given by

$$f(x) = P(X = x) = (1-p)^{x-1} p.$$

Hence, the probability density function of X is

$$f(x) = (1-p)^{x-1} p \quad x = 1, 2, 3, \dots, \infty,$$

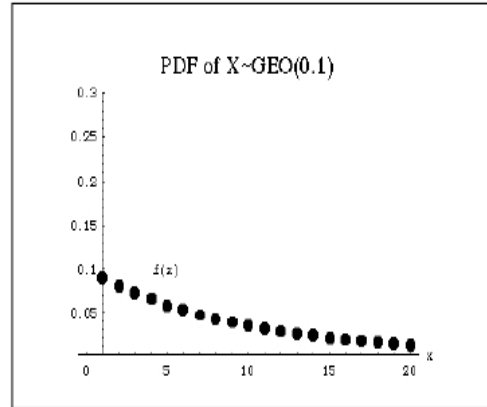
where p denotes the probability of success in a single Bernoulli trial.

Definition A random variable X has a geometric distribution if its probability density function is given by

$$f(x) = (1 - p)^{x-1} p \quad x = 1, 2, 3, \dots, \infty,$$

where p denotes the probability of success in a single Bernoulli trial.

If X has a geometric distribution we denote it as $X \sim GEO(p)$.



Theorem If X is a geometric random variable with parameter p , then the mean, variance and moment generating functions are respectively given by

$$\begin{aligned} \mu_X &= \frac{1}{p} \\ \sigma_X^2 &= \frac{1-p}{p^2} \\ M_X(t) &= \frac{pe^t}{1 - (1-p)e^t}, \quad \text{if } t < -\ln(1-p). \end{aligned}$$

Negative Binomial Distribution

Let X denote the trial number on which the r^{th} success occurs. Here r is a positive integer greater than or equal to one. This is equivalent to saying that the random variable X denotes the number of trials needed to observe the r^{th} successes. Suppose we want to find the probability that the fifth head is observed on the 10th independent flip of an unbiased coin. This is a case of finding $P(X = 10)$. Let us find the general case $P(X = x)$.

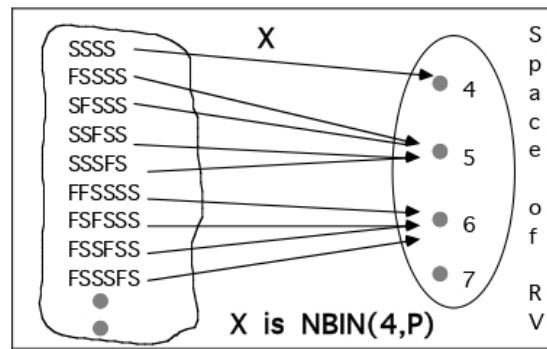
$$\begin{aligned}
 P(X = x) &= P(\text{first } x - 1 \text{ trials contain } x - r \text{ failures and } r - 1 \text{ successes}) \\
 &\quad \cdot P(r^{\text{th}} \text{ success in } x^{\text{th}} \text{ trial}) \\
 &= \binom{x-1}{r-1} p^{r-1} (1-p)^{x-r} p \\
 &= \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots, \infty.
 \end{aligned}$$

Hence the probability density function of the random variable X is given by

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots, \infty.$$

Notice that this probability density function $f(x)$ can also be expressed as

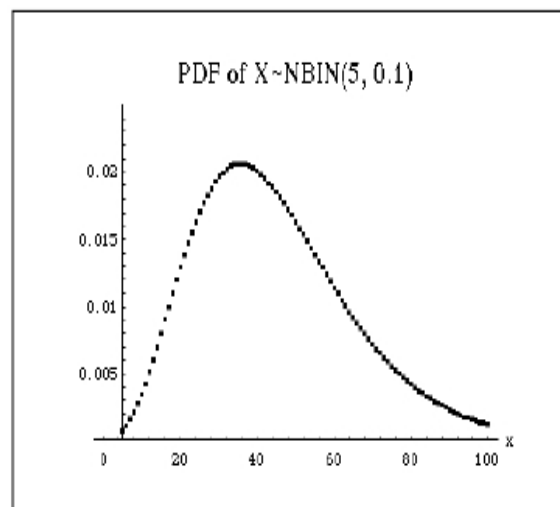
$$f(x) = \binom{x+r-1}{r-1} p^r (1-p)^x, \quad x = 0, 1, \dots, \infty.$$



Definition A random variable X has the negative binomial (or Pascal) distribution if its probability density function is of the form

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \dots, \infty,$$

where p is the probability of success in a single Bernoulli trial. We denote the random variable X whose distribution is negative binomial distribution by writing $X \sim \text{NBIN}(r, p)$.



We need the following technical result to show that the above function is really a probability density function. The technical result we are going to establish is called the negative binomial theorem.

Theorem If $X \sim NBIN(r, p)$, then

$$E(X) = \frac{r}{p}$$

$$Var(X) = \frac{r(1-p)}{p^2}$$

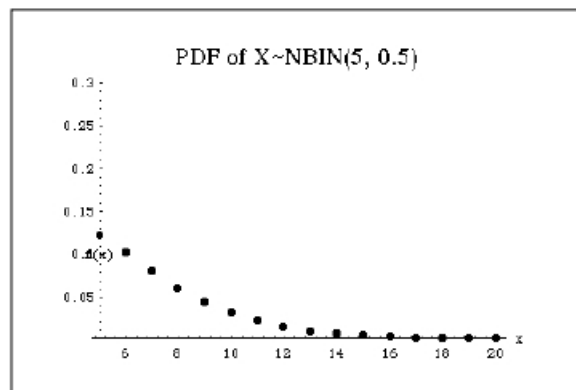
$$M(t) = \left(\frac{pe^t}{1 - (1-p)e^t} \right)^r, \quad \text{if } t < -\ln(1-p).$$

Example What is the probability that the fifth head is observed on the 10th independent flip of a coin?

Answer: Let X denote the number of trials needed to observe 5th head. Hence X has a negative binomial distribution with $r = 5$ and $p = \frac{1}{2}$.

We want to find

$$\begin{aligned} P(X = 10) &= f(10) \\ &= \binom{9}{4} p^5 (1-p)^5 \\ &= \binom{9}{4} \left(\frac{1}{2}\right)^{10} \\ &= \frac{63}{512}. \end{aligned}$$



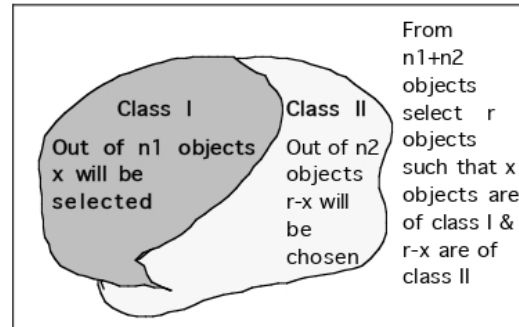
Hypergeometric Distribution

Consider a collection of n objects which can be classified into two classes, say class 1 and class 2. Suppose that there are n_1 objects in class 1 and n_2 objects in class 2. A collection of r objects is selected from these n objects at random and without replacement. We are interested in finding out the probability that exactly x of these r objects are from class 1. If x of these r objects are from class 1, then the remaining $r - x$ objects must be from class 2. We can select x objects from class 1 in any one of $\binom{n_1}{x}$ ways. Similarly, the remaining $r - x$ objects can be selected in $\binom{n_2}{r-x}$ ways. Thus, the number of ways one can select a subset of r objects from a set of n objects, such that

x number of objects will be from class 1 and $r - x$ number of objects will be from class 2, is given by $\binom{n_1}{x} \binom{n_2}{r-x}$. Hence,

$$P(X = x) = \frac{\binom{n_1}{x} \binom{n_2}{r-x}}{\binom{n}{r}},$$

where $x \leq r$, $x \leq n_1$ and $r - x \leq n_2$.



Definition A random variable X is said to have a hypergeometric distribution if its probability density function is of the form

$$f(x) = \frac{\binom{n_1}{x} \binom{n_2}{r-x}}{\binom{n_1+n_2}{r}}, \quad x = 0, 1, 2, \dots, r$$

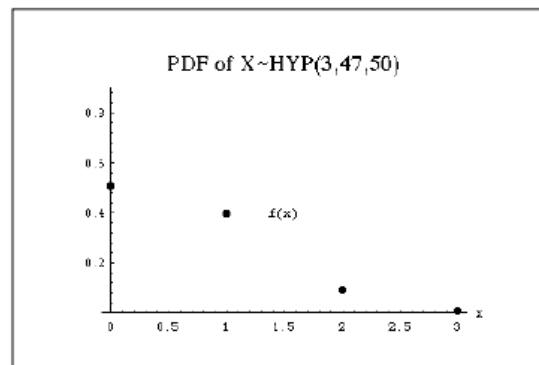
where $x \leq n_1$ and $r - x \leq n_2$ with n_1 and n_2 being two positive integers. We shall denote such a random variable by writing

$$X \sim HYP(n_1, n_2, r).$$

Example Suppose there are 3 defective items in a lot of 50 items. A sample of size 10 is taken at random and without replacement. Let X denote the number of defective items in the sample. What is the probability that the sample contains at most one defective item?

Answer: Clearly, $X \sim HYP(3, 47, 10)$. Hence the probability that the sample contains at most one defective item is

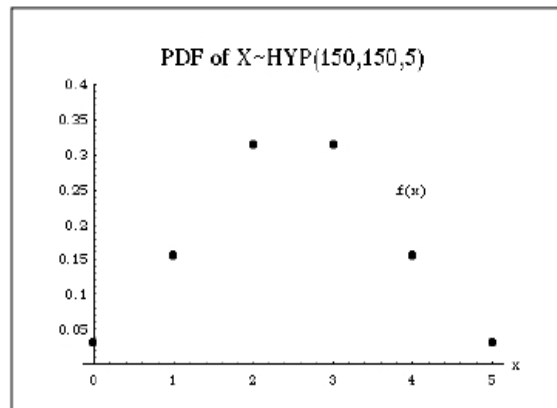
$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= \frac{\binom{3}{0} \binom{47}{10}}{\binom{50}{10}} + \frac{\binom{3}{1} \binom{47}{9}}{\binom{50}{10}} \\ &= 0.504 + 0.4 \\ &= 0.904. \end{aligned}$$



Example A random sample of 5 students is drawn without replacement from among 300 seniors, and each of these 5 seniors is asked if she/he has tried a certain drug. Suppose 50% of the seniors actually have tried the drug. What is the probability that two of the students interviewed have tried the drug?

Answer: Let X denote the number of students interviewed who have tried the drug. Hence the probability that two of the students interviewed have tried the drug is

$$\begin{aligned} P(X = 2) &= \frac{\binom{150}{2} \binom{150}{3}}{\binom{300}{5}} \\ &= 0.3146. \end{aligned}$$



Example A radio supply house has 200 transistor radios, of which 3 are improperly soldered and 197 are properly soldered. The supply house randomly draws 4 radios without replacement and sends them to a customer. What is the probability that the supply house sends 2 improperly soldered radios to its customer?

Answer: The probability that the supply house sends 2 improperly soldered

radios to its customer is

$$P(X = 2) = \frac{\binom{3}{2} \binom{197}{2}}{\binom{200}{4}} = 0.000895.$$

Theorem If $X \sim HYP(n_1, n_2, r)$, then

$$E(X) = r \frac{n_1}{n_1 + n_2}$$

$$Var(X) = r \left(\frac{n_1}{n_1 + n_2} \right) \left(\frac{n_2}{n_1 + n_2} \right) \left(\frac{n_1 + n_2 - r}{n_1 + n_2 - 1} \right).$$

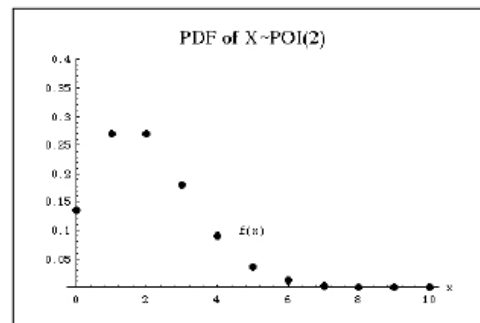
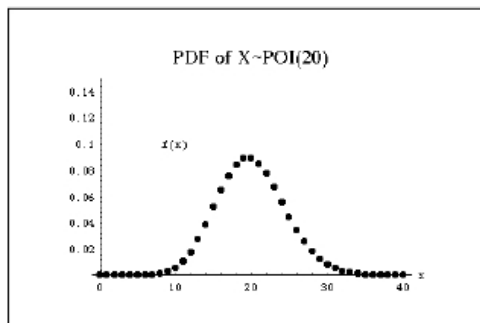
Poisson Distribution

In this section, we define an important discrete distribution which is widely used for modeling many real life situations. First, we define this distribution and then we present some of its important properties.

Definition A random variable X is said to have a Poisson distribution if its probability density function is given by

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty,$$

where $0 < \lambda < \infty$ is a parameter. We denote such a random variable by $X \sim \text{POI}(\lambda)$.



The probability density function f is called the Poisson distribution after Simeon D. Poisson (1781-1840).

Example Is the real valued function defined by

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots, \infty,$$

where $0 < \lambda < \infty$ is a parameter, a probability density function?

Answer: It is easy to check $f(x) \geq 0$. We show that $\sum_{x=0}^{\infty} f(x)$ is equal to one.

$$\begin{aligned}\sum_{x=0}^{\infty} f(x) &= \sum_{x=0}^{\infty} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} e^{\lambda} = 1.\end{aligned}$$

Theorem If $X \sim POI(\lambda)$, then

$$\begin{aligned}E(X) &= \lambda \\ Var(X) &= \lambda \\ M(t) &= e^{\lambda(e^t-1)}.\end{aligned}$$

Proof: First, we find the moment generating function of X .

$$\begin{aligned}M(t) &= \sum_{x=0}^{\infty} e^{tx} f(x) \\ &= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} e^{tx} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^t \lambda)^x}{x!} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= e^{\lambda(e^t-1)}.\end{aligned}$$

Thus,

$$M'(t) = \lambda e^t e^{\lambda(e^t-1)},$$

and

$$E(X) = M'(0) = \lambda.$$

Similarly,

$$M''(t) = \lambda e^t e^{\lambda(e^t-1)} + (\lambda e^t)^2 e^{\lambda(e^t-1)}.$$

Hence

$$M''(0) = E(X^2) = \lambda^2 + \lambda.$$

Therefore

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

This completes the proof.

Example A random variable X has a Poisson distribution with a mean of 3. What is the probability that X is bounded by 1 and 3, that is, $P(1 \leq X \leq 3)$?

Answer:

$$\mu_X = 3 = \lambda$$

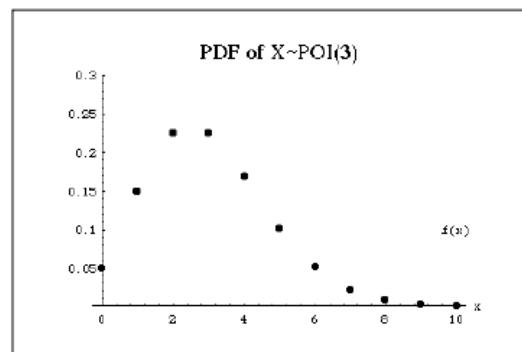
$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

Hence

$$f(x) = \frac{3^x e^{-3}}{x!}, \quad x = 0, 1, 2, \dots$$

Therefore

$$\begin{aligned} P(1 \leq X \leq 3) &= f(1) + f(2) + f(3) \\ &= 3e^{-3} + \frac{9}{2}e^{-3} + \frac{27}{6}e^{-3} \\ &= 12e^{-3}. \end{aligned}$$



Example The number of traffic accidents per week in a small city has a Poisson distribution with mean equal to 3. What is the probability of exactly 2 accidents occur in 2 weeks?

Answer: The mean traffic accident is 3. Thus, the mean accidents in two weeks are

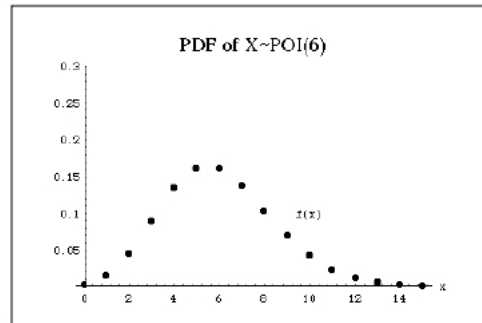
$$\lambda = (3)(2) = 6.$$

Since

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

we get

$$f(2) = \frac{6^2 e^{-6}}{2!} = 18e^{-6}.$$



Example Let X have a Poisson distribution with parameter $\lambda = 1$. What is the probability that $X \geq 2$ given that $X \leq 4$?

Answer:

$$P(X \geq 2 / X \leq 4) = \frac{P(2 \leq X \leq 4)}{P(X \leq 4)}.$$

$$\begin{aligned} P(2 \leq X \leq 4) &= \sum_{x=2}^4 \frac{\lambda^x e^{-\lambda}}{x!} \\ &= \frac{1}{e} \sum_{x=2}^4 \frac{1}{x!} \\ &= \frac{17}{24e}. \end{aligned}$$

Similarly

$$\begin{aligned} P(X \leq 4) &= \frac{1}{e} \sum_{x=0}^4 \frac{1}{x!} \\ &= \frac{65}{24e}. \end{aligned}$$

Therefore, we have

$$P(X \geq 2 / X \leq 4) = \frac{17}{65}.$$

SOME SPECIAL CONTINUOUS DISTRIBUTIONS

In this chapter, we study some well known continuous probability density functions. We want to study them because they arise in many applications. We begin with the simplest probability density function.

Uniform Distribution

Let the random variable X denote the outcome when a point is selected at random from an interval $[a, b]$. We want to find the probability of the event $X \leq x$, that is we would like to determine the probability that the point selected from $[a, b]$ would be less than or equal to x . To compute this probability, we need a probability measure μ that satisfies the three axioms of Kolmogorov (namely nonnegativity, normalization and countable additivity). For continuous variables, the events are interval or union of intervals. The length of the interval when normalized satisfies all the three axioms and thus it can be used as a probability measure for one-dimensional random variables. Hence

$$P(X \leq x) = \frac{\text{length of } [a, x]}{\text{length of } [a, b]}.$$

Thus, the cumulative distribution function F is

$$F(x) = P(X \leq x) = \frac{x - a}{b - a}, \quad a \leq x \leq b,$$

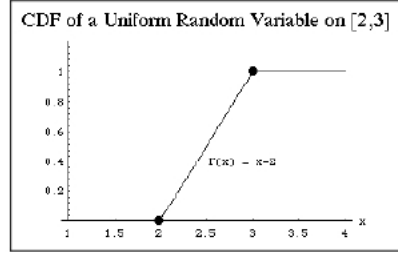
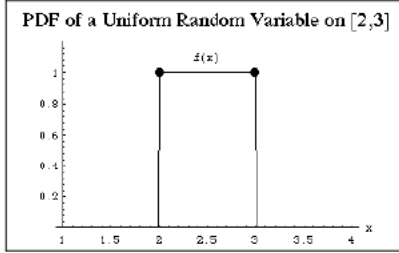
where a and b are any two real constants with $a < b$. To determine the probability density function from cumulative density function, we calculate the derivative of $F(x)$. Hence

$$f(x) = \frac{d}{dx}F(x) = \frac{1}{b - a}, \quad a \leq x \leq b.$$

Definition A random variable X is said to be uniform on the interval $[a, b]$ if its probability density function is of the form

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b,$$

where a and b are constants. We denote a random variable X with the uniform distribution on the interval $[a, b]$ as $X \sim UNIF(a, b)$.



The uniform distribution provides a probability model for selecting points at random from an interval $[a, b]$. An important application of uniform distribution lies in random number generation. The following theorem gives the mean, variance and moment generating function of a uniform random variable.

Theorem If X is uniform on the interval $[a, b]$ then the mean, variance and moment generating function of X are given by

$$\begin{aligned} E(X) &= \frac{b+a}{2} \\ \text{Var}(X) &= \frac{(b-a)^2}{12} \\ M(t) &= \begin{cases} 1 & \text{if } t = 0 \\ \frac{e^{tb} - e^{ta}}{t(b-a)}, & \text{if } t \neq 0 \end{cases} \end{aligned}$$

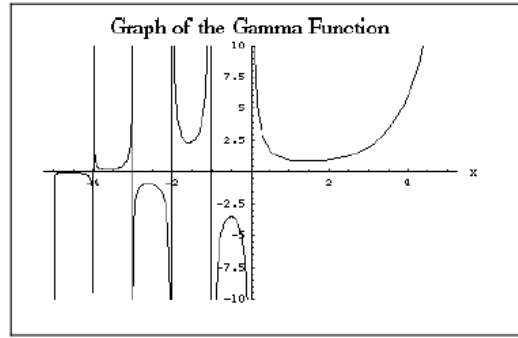
Gamma Distribution

The gamma distribution involves the notion of gamma function. First, we develop the notion of gamma function and study some of its well known properties. The gamma function, $\Gamma(z)$, is a generalization of the notion of factorial. The gamma function is defined as

$$\Gamma(z) := \int_0^{\infty} x^{z-1} e^{-x} dx,$$

where z is positive real number (that is, $z > 0$). The condition $z > 0$ is assumed for the convergence of the integral. Although the integral does not converge for $z < 0$, it can be shown by using an alternative definition of gamma function that it is defined for all $z \in \mathbb{R} \setminus \{0, -1, -2, -3, \dots\}$.

The integral on the right side of the above expression is called Euler's second integral, after the Swiss mathematician Leonhard Euler (1707-1783). The graph of the gamma function is shown below. Observe that the zero and negative integers correspond to vertical asymptotes of the graph of gamma function.



Lemma $\Gamma(1) = 1$.

Proof:

$$\Gamma(1) = \int_0^{\infty} x^0 e^{-x} dx = [-e^{-x}]_0^{\infty} = 1.$$

Lemma The gamma function $\Gamma(z)$ satisfies the functional equation $\Gamma(z) = (z-1)\Gamma(z-1)$ for all real number $z > 1$.

Proof: Let z be a real number such that $z > 1$, and consider

$$\begin{aligned}\Gamma(z) &= \int_0^\infty x^{z-1} e^{-x} dx \\ &= [-x^{z-1} e^{-x}]_0^\infty + \int_0^\infty (z-1) x^{z-2} e^{-x} dx \\ &= (z-1) \int_0^\infty x^{z-2} e^{-x} dx \\ &= (z-1) \Gamma(z-1).\end{aligned}$$

Although, we have proved this lemma for all real $z > 1$, actually this lemma holds also for all real number $z \in \mathbb{R} \setminus \{1, 0, -1, -2, -3, \dots\}$.

Lemma $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

Proof: We want to show that

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx$$

is equal to $\sqrt{\pi}$. We substitute $y = \sqrt{x}$, hence the above integral becomes

$$\begin{aligned}\Gamma\left(\frac{1}{2}\right) &= \int_0^\infty \frac{e^{-x}}{\sqrt{x}} dx \\ &= 2 \int_0^\infty e^{-y^2} dy, \quad \text{where } y = \sqrt{x}.\end{aligned}$$

Hence

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^\infty e^{-u^2} du$$

and also

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^\infty e^{-v^2} dv.$$

Multiplying the above two expressions, we get

$$\left(\Gamma\left(\frac{1}{2}\right)\right)^2 = 4 \int_0^\infty \int_0^\infty e^{-(u^2+v^2)} du dv.$$

Now we change the integral into polar form by the transformation $u = r \cos(\theta)$ and $v = r \sin(\theta)$. The Jacobian of the transformation is

$$\begin{aligned}J(r, \theta) &= \det \begin{pmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial \theta} \end{pmatrix} \\ &= \det \begin{pmatrix} \cos(\theta) & -r \sin(\theta) \\ \sin(\theta) & r \cos(\theta) \end{pmatrix} \\ &= r \cos^2(\theta) + r \sin^2(\theta) \\ &= r.\end{aligned}$$

Hence, we get

$$\begin{aligned}
 \left(\Gamma \left(\frac{1}{2} \right) \right)^2 &= 4 \int_0^{\frac{\pi}{2}} \int_0^{\infty} e^{-r^2} J(r, \theta) dr d\theta \\
 &= 4 \int_0^{\frac{\pi}{2}} \int_0^{\infty} e^{-r^2} r dr d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \int_0^{\infty} e^{-r^2} 2r dr d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \left[\int_0^{\infty} e^{-r^2} dr^2 \right] d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \Gamma(1) d\theta \\
 &= \pi.
 \end{aligned}$$

Therefore, we get

$$\Gamma \left(\frac{1}{2} \right) = \sqrt{\pi}.$$

Example If n is a natural number, then $\Gamma(n+1) = n!$.

Answer:

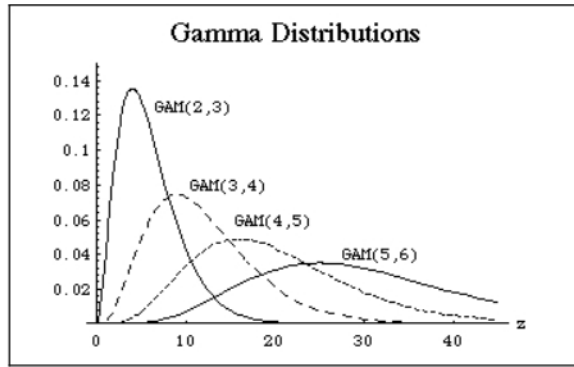
$$\begin{aligned}
 \Gamma(n+1) &= n \Gamma(n) \\
 &= n(n-1) \Gamma(n-1) \\
 &= n(n-1)(n-2) \Gamma(n-2) \\
 &= \dots \dots \\
 &= n(n-1)(n-2) \dots (1) \Gamma(1) \\
 &= n!
 \end{aligned}$$

Now we are ready to define the gamma distribution.

Definition A continuous random variable X is said to have a gamma distribution if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} & \text{if } 0 < x < \infty \\ 0 & \text{otherwise,} \end{cases}$$

where $\alpha > 0$ and $\theta > 0$. We denote a random variable with gamma distribution as $X \sim GAM(\theta, \alpha)$. The following diagram shows the graph of the gamma density for various values of values of the parameters θ and α .



The following theorem gives the expected value, the variance, and the moment generating function of the gamma random variable

Theorem . If $X \sim GAM(\theta, \alpha)$, then

$$E(X) = \theta \alpha$$

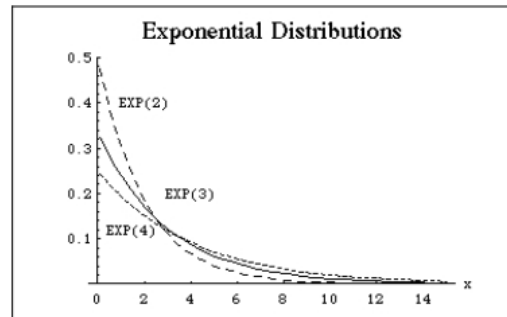
$$Var(X) = \theta^2 \alpha$$

$$M(t) = \left(\frac{1}{1 - \theta t} \right)^\alpha, \quad \text{if } t < \frac{1}{\theta}.$$

Definition A continuous random variable is said to be an exponential random variable with parameter θ if its probability density function is of the form

$$f(x) = \begin{cases} \frac{1}{\theta} e^{-\frac{x}{\theta}} & \text{if } x > 0 \\ 0 & \text{otherwise,} \end{cases}$$

where $\theta > 0$. If a random variable X has an exponential density function with parameter θ , then we denote it by writing $X \sim EXP(\theta)$.



An exponential distribution is a special case of the gamma distribution. If the parameter $\alpha = 1$, then the gamma distribution reduces to the exponential distribution. Hence most of the information about an exponential distribution can be obtained from the gamma distribution.

Beta Distribution

The beta distribution is one of the basic distributions in statistics. It has many applications in classical as well as Bayesian statistics. It is a versatile distribution and as such it is used in modeling the behavior of random variables that are positive but bounded in possible values. Proportions and percentages fall in this category.

The beta distribution involves the notion of beta function. First we explain the notion of the beta integral and some of its simple properties. Let α and β be any two positive real numbers. The beta function $B(\alpha, \beta)$ is defined as

$$B(\alpha, \beta) = \int_0^1 x^{\alpha-1} (1-x)^{\beta-1} dx.$$

First, we prove a theorem that establishes the connection between the beta function and the gamma function.

Theorem Let α and β be any two positive real numbers. Then

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)},$$

where

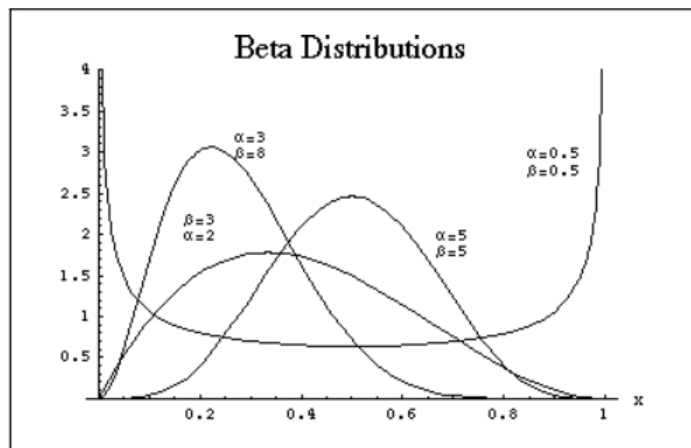
$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x} dx$$

Definition A random variable X is said to have the beta density function if its probability density function is of the form

$$f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

for every positive α and β . If X has a beta distribution, then we symbolically denote this by writing $X \sim BETA(\alpha, \beta)$.

The following figure illustrates the graph of the beta distribution for various values of α and β .



The beta distribution reduces to the uniform distribution over $(0, 1)$ if $\alpha = 1 = \beta$. The following theorem gives the mean and variance of the beta distribution.

Theorem If $X \sim BETA(\alpha, \beta)$,

$$E(X) = \frac{\alpha}{\alpha + \beta}$$

$$Var(X) = \frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}.$$

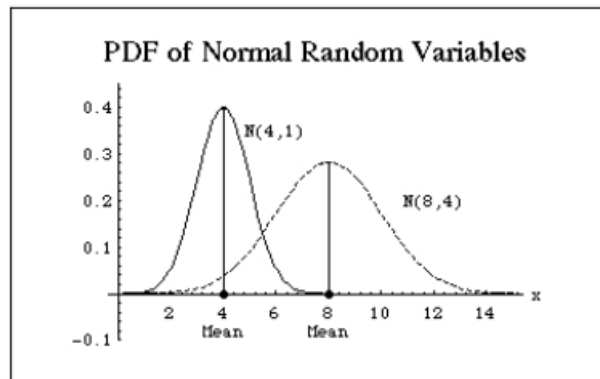
Normal Distribution

Among continuous probability distributions, the normal distribution is very well known since it arises in many applications. Normal distribution was discovered by a French mathematician Abraham DeMoivre (1667-1754). DeMoivre wrote two important books. One is called the *Annuities Upon Lives*, the first book on actuarial sciences and the second book is called the *Doctrine of Chances*, one of the early books on the probability theory. Pierre-Simon Laplace (1749-1827) applied normal distribution to astronomy. Carl Friedrich Gauss (1777-1855) used normal distribution in his studies of problems in physics and astronomy. Adolphe Quetelet (1796-1874) demonstrated that man's physical traits (such as height, chest expansion, weight etc.) as well as social traits follow normal distribution. The main importance of normal distribution lies on the central limit theorem which says that the sample mean has a normal distribution if the sample size is large.

Definition A random variable X is said to have a normal distribution if its probability density function is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty,$$

where $-\infty < \mu < \infty$ and $0 < \sigma^2 < \infty$ are arbitrary parameters. If X has a normal distribution with parameters μ and σ^2 , then we write $X \sim N(\mu, \sigma^2)$.



Example Is the real valued function defined by

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$

a probability density function of some random variable X ?

Answer: To answer this question, we must check that f is nonnegative and it integrates to 1. The nonnegative part is trivial since the exponential function is always positive. Hence using property of the gamma function, we show that f integrates to 1 on $\mathbb{R}$.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_{-\infty}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= 2 \int_{\mu}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2} dx \\ &= \frac{2}{\sigma \sqrt{2\pi}} \int_0^{\infty} e^{-z} \frac{\sigma}{\sqrt{2z}} dz, \quad \text{where } z = \frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2 \\ &= \frac{1}{\sqrt{\pi}} \int_0^{\infty} \frac{1}{\sqrt{z}} e^{-z} dz \\ &= \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = \frac{1}{\sqrt{\pi}} \sqrt{\pi} = 1. \end{aligned}$$

Theorem If $X \sim N(\mu, \sigma^2)$, then

$$\begin{aligned} E(X) &= \mu \\ \text{Var}(X) &= \sigma^2 \\ M(t) &= e^{\mu t + \frac{1}{2} \sigma^2 t^2}. \end{aligned}$$

Bibliography

Prasanna Sahoo, “*Probability and Mathematical Statistics*”, January 2015.

Mark L. Berenson, David M. Levine, Timothy C. Krehbiel, “*Basic Business Statistics: Concepts and Applications*”, 12th edition, PEARSON.